

# CAVY

## Application walkthrough: every screen and dialog

CAVY (Cognitive Interaction Quotient) is a classroom platform for learning to code. It records how students write, run and fix code and how they use an AI assistant, then turns that into a score for each session and a view of progress over time.

This document follows the application in the order a classroom would meet it: connecting, a teacher preparing a class, a student working, the scores, and finally the administrator's panel. Each page is one screen, named and explained.

Version 0.1.0 · captured 03 October 2026 · 74 screens

Captured from the running application with demonstration data: one administrator, two teachers, three students. The operating system's own file chooser and Save dialogs are not shown.

# Who uses what

## TEACHER

### **Prepares and reviews**

1. Signs in and builds a class
2. Publishes a lab with three stages
3. Shares files, links, instructions
4. Watches submissions arrive live
5. Reads reports, adds follow-ups

## STUDENT

### **Learns and is scored**

1. Signs in; sees the teacher's labs
2. Works through Learning, Exploration, Assessment
3. Writes and runs code, may ask the AI
4. Explains the solution; gets a CIQ score
5. Tracks progress over time

## ADMINISTRATOR

### **Runs the classroom**

1. Sees who is online and working
2. Manages and assigns accounts
3. Signs people out, resets passwords
4. Browses the data, reads the audit log
5. Sets the AI assistant, takes backups

**One server holds everything.** Teachers, students and the administrator each use the CAVY app on their own computer; all accounts, labs, work and scores are stored on the one computer that hosts the server, and screens update live without refreshing. A student's code runs on the student's own computer.

# Contents

## 1 • Getting connected

Opening the app, connecting to a classroom server, and creating accounts.

|                                           |    |
|-------------------------------------------|----|
| Login screen (Student)                    | 6  |
| Login screen (Teacher tab)                | 7  |
| Connect to a server (dialog)              | 8  |
| Connect to a server (unreachable address) | 9  |
| Login screen (connected to the server)    | 10 |
| Host a classroom server (dialog)          | 11 |
| Hosting a server (running)                | 12 |
| Create an account (Teacher)               | 13 |
| Create an account (Student)               | 14 |
| Create an account (filled in)             | 15 |

## 2 • The teacher sets up a class

A teacher signs in, builds a class and publishes a lab.

|                                     |    |
|-------------------------------------|----|
| AI assistant setup (Groq)           | 17 |
| AI assistant setup (Ollama)         | 18 |
| After signing in (My Labs)          | 19 |
| My Class (before adding students)   | 20 |
| My Class (choosing students to add) | 21 |
| My Class (after adding)             | 22 |
| Create New Session (empty form)     | 23 |
| Create New Session (filled in, top) | 24 |
| Create New Session (stage settings) | 25 |
| My Labs (new lab published)         | 26 |

## 3 • The teacher shares resources

Files, links and written instructions, for the whole class or chosen students, attached to labs.

|                                              |    |
|----------------------------------------------|----|
| Teacher Resources (empty)                    | 28 |
| Add resource: File                           | 29 |
| Add resource: File (filled in)               | 30 |
| Add resource: Link                           | 31 |
| Add resource: Instructions (chosen students) | 32 |
| Teacher Resources (three items)              | 33 |
| Edit resource                                | 34 |

## 4 • The student starts a lab

What a student sees when they sign in and open a lab.

|                                     |    |
|-------------------------------------|----|
| Student Home                        | 36 |
| Student Home (lab sessions)         | 37 |
| Lab stages                          | 38 |
| Lab stages (resources for this lab) | 39 |

# Contents (continued)

## 5 • The coding workspace

Writing and running code, asking the AI assistant, and the restricted Assessment stage.

|                                         |    |
|-----------------------------------------|----|
| Workspace (start of the Learning stage) | 41 |
| Workspace (writing code)                | 42 |
| Workspace (running code)                | 43 |
| AI assistant (question typed)           | 44 |
| AI assistant (answer)                   | 45 |
| Assessment stage (AI restricted)        | 46 |
| Practice workspace                      | 47 |

## 6 • Submitting and scores

Submitting work, explaining a solution, and the Cognitive Interaction Quotient (CIQ) score.

|                                            |    |
|--------------------------------------------|----|
| Submission confirmation                    | 49 |
| CIQ score (Learning stage)                 | 50 |
| Concept check (dialog)                     | 51 |
| Submission confirmation (Assessment)       | 52 |
| CIQ score (Assessment, with concept check) | 53 |

## 7 • The student's own pages

Resources, profile and progress over time.

|                                     |    |
|-------------------------------------|----|
| Student Resources                   | 55 |
| Reading written instructions        | 56 |
| Student Profile                     | 57 |
| My Progress (overview)              | 58 |
| My Progress (strengths and history) | 59 |

## 8 • The teacher reviews and manages

Live dashboard, reports, follow-up assessments, editing labs and helping students.

|                                           |    |
|-------------------------------------------|----|
| Teacher Home (dashboard)                  | 61 |
| Reports (all labs)                        | 62 |
| Lab report                                | 63 |
| Add a follow-up assessment (dialog)       | 64 |
| Edit Session                              | 65 |
| Archive a lab (confirmation)              | 66 |
| My Class (three students)                 | 67 |
| Reset a student's password (confirmation) | 68 |
| Temporary password (shown once)           | 69 |
| Teacher Profile                           | 70 |

## 9 • The administrator panel

Everything the person running the server can see and control.

|                                          |    |
|------------------------------------------|----|
| Admin: first-time setup                  | 72 |
| Admin: sign in                           | 73 |
| Admin: Overview (live)                   | 74 |
| Admin: Overview (activity)               | 75 |
| Admin: Users                             | 76 |
| Admin: Edit an account                   | 77 |
| Admin: Disable an account (confirmation) | 78 |
| Admin: Labs                              | 79 |
| Admin: Resources                         | 80 |
| Admin: Database browser                  | 81 |
| Admin: Database row                      | 82 |
| Admin: One session in detail             | 83 |
| Admin: Audit log                         | 84 |
| Admin: AI assistant                      | 85 |
| Admin: Backup                            | 86 |
| Student signed out by the administrator  | 87 |

# 1 • Getting connected

Opening the app, connecting to a classroom server, and creating accounts.

# Login screen (Student)

**CAVY**  
COGNITIVE INTERACTION QUOTIENT

## Collaboration has a score now.

CAVY doesn't just grade what you submit — it measures your CIQ: how well you question, edit, and reason alongside AI, session by session.

**3**  
Pillars measured

**14**  
Behavioral signals



### Welcome back

Sign in to pick up your session where you left it.

Student

Teacher

Email

you@school.edu

Password

.....

Log in

New to CAVY? [Create an account](#)

Connected to <http://127.0.0.1:8000> · [Change server](#)  
[Host a server on this computer](#)

The first screen. Students and teachers sign in with an email and password. The line at the bottom shows whether this computer is connected to a classroom server.

## Login screen (Teacher tab)

**CAVY**  
COGNITIVE INTERACTION QUOTIENT

### Collaboration has a score now.

CAVY doesn't just grade what you submit — it measures your CIQ: how well you question, edit, and reason alongside AI, session by session.

3  
Pillars measured

14  
Behavioral signals



### Welcome back

Sign in to pick up your session where you left it.

Student

Teacher

Email

you@school.edu

Password

.....

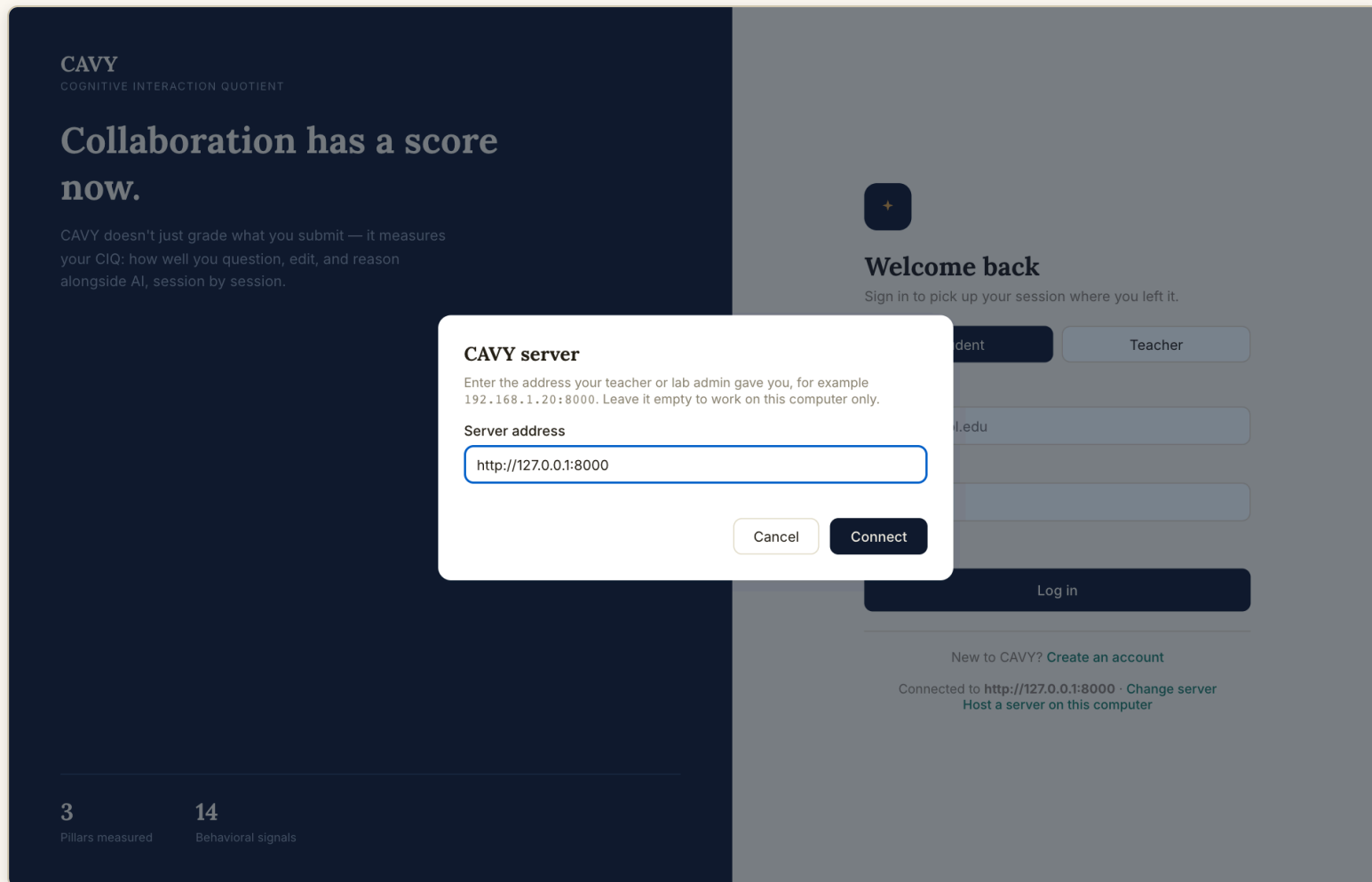
Log in

New to CAVY? [Create an account](#)

Connected to <http://127.0.0.1:8000> · [Change server](#)  
[Host a server on this computer](#)

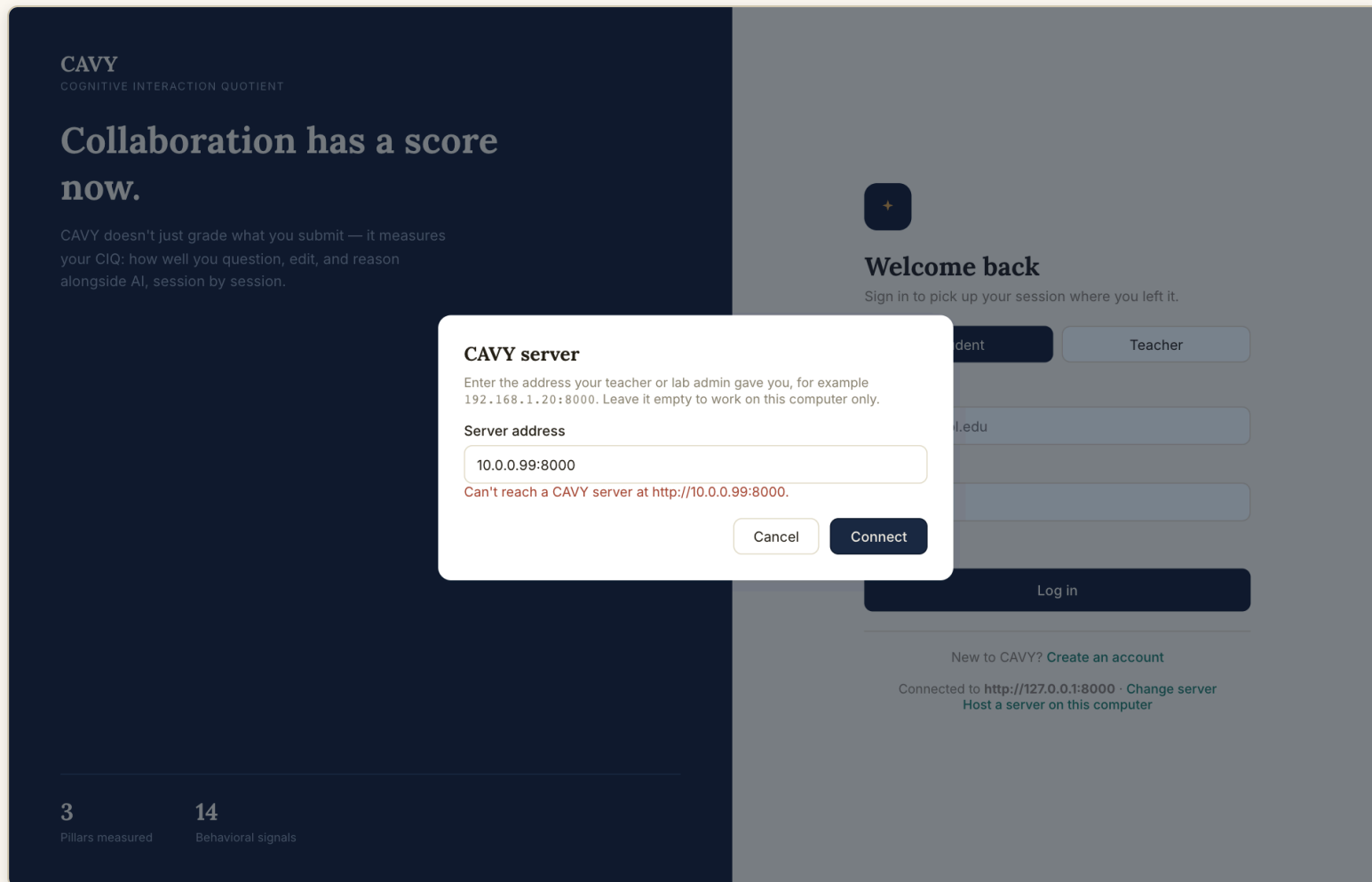
Teachers use the Teacher tab. The same screen serves both roles.

## Connect to a server (dialog)



'Connect to a server' asks for the address the classroom host gives out. The address is checked before it is saved.

## Connect to a server (unreachable address)



A wrong or unreachable address is reported clearly instead of failing silently, and nothing is changed.

## Login screen (connected to the server)

**CAVY**  
COGNITIVE INTERACTION QUOTIENT

### Collaboration has a score now.

CAVY doesn't just grade what you submit — it measures your CIQ: how well you question, edit, and reason alongside AI, session by session.

3  
Pillars measured

14  
Behavioral signals



### Welcome back

Sign in to pick up your session where you left it.

Student

Teacher

Email

you@school.edu

Password

.....

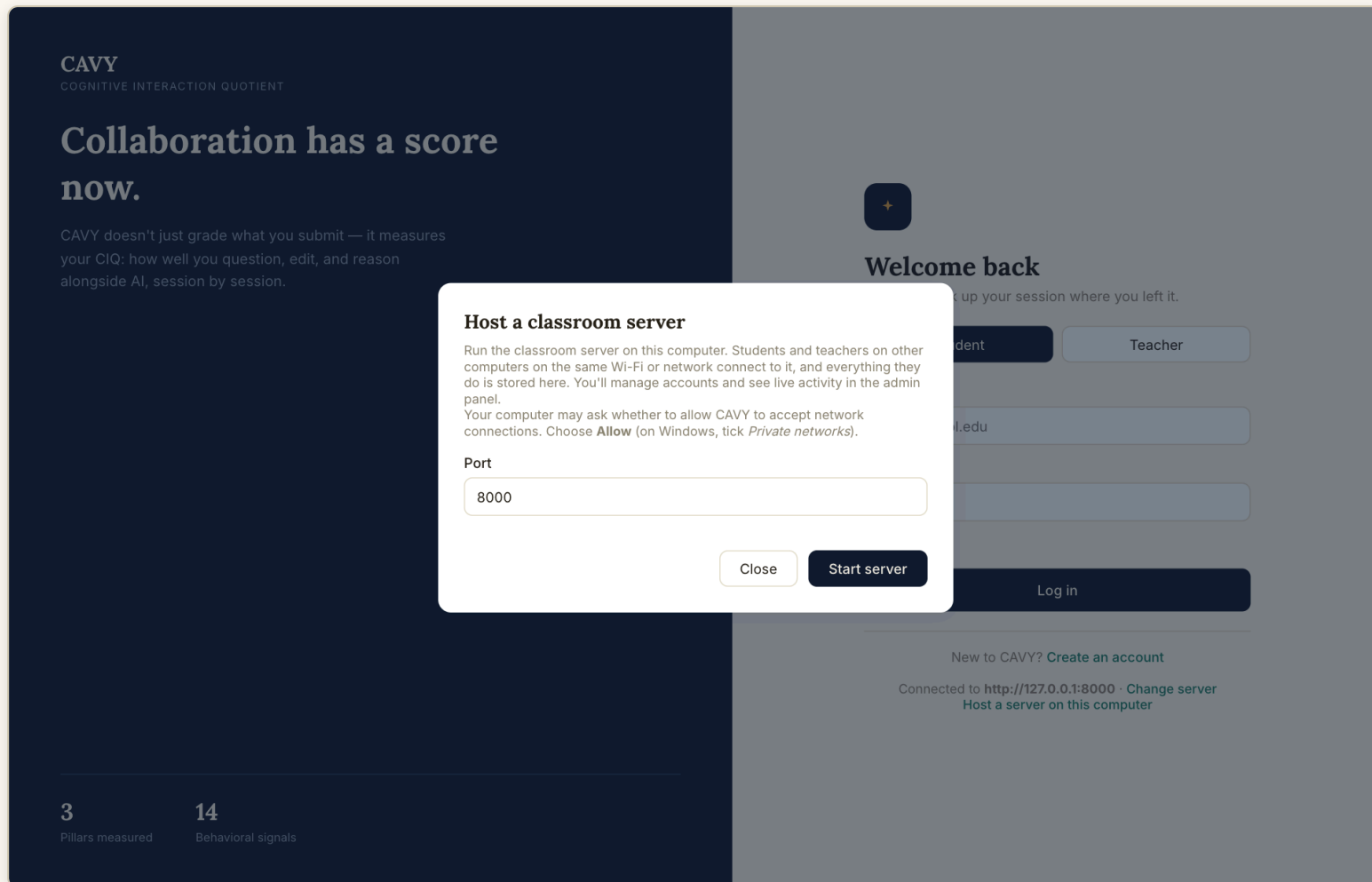
Log in

New to CAVY? [Create an account](#)

Connected to <http://127.0.0.1:8000> · [Change server](#)  
[Host a server on this computer](#)

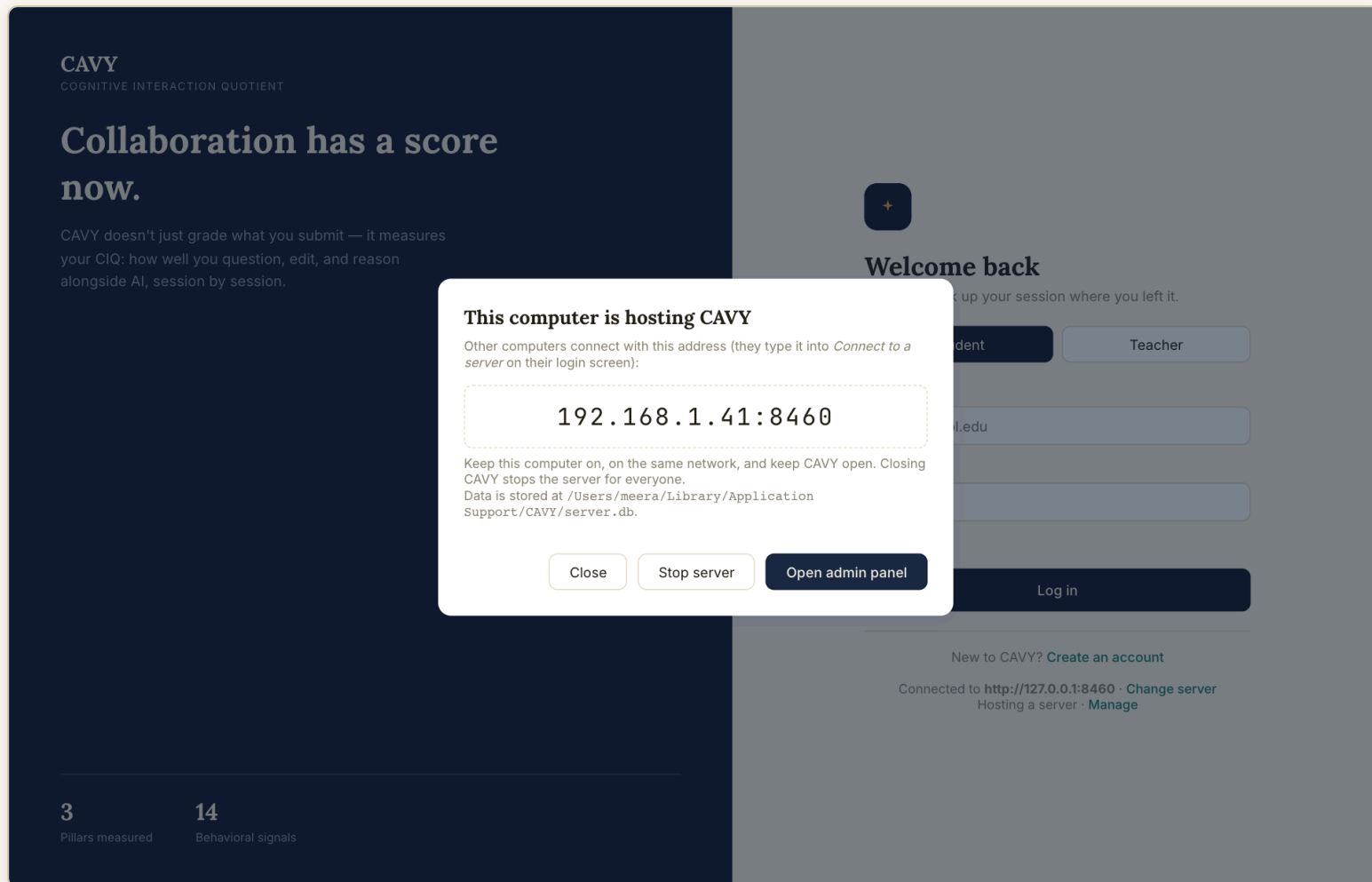
Once connected, the bottom line shows the server address. The app remembers it for next time.

## Host a classroom server (dialog)



One computer can host the classroom server straight from the app, with no terminal. Everyone else connects to it.

## Hosting a server (running)



While hosting, the dialog shows the address other computers type in, and a button that opens the admin panel.

## Create an account (Teacher)

**CAVY**  
COGNITIVE INTERACTION QUOTIENT

### Collaboration has a score now.

CAVY doesn't just grade what you submit — it measures your CIQ: how well you question, edit, and reason alongside AI, session by session.

3  
Pillars measured

14  
Behavioral signals



### Create an account

Set up your CAVY login.

Student

Teacher

**Full Name**

**Email**

**Password**

Create Account

Already have an account? [Log in](#)

Teachers create an account the same way. A teacher account can then publish labs, add students to a class and share resources.

## Create an account (Student)

**CAVY**  
COGNITIVE INTERACTION QUOTIENT

### Collaboration has a score now.

CAVY doesn't just grade what you submit — it measures your CIQ: how well you question, edit, and reason alongside AI, session by session.

3  
Pillars measured

14  
Behavioral signals

### Create an account

Set up your CAVY login.

Student

Teacher

**Full Name**

**Email**

**Password**

**Enrolment No.**  
(optional)

Create Account

[Already have an account? Log in](#)

New students create their own account. The enrolment number is optional. Passwords need at least 8 characters.

## Create an account (filled in)

**CAVY**  
COGNITIVE INTERACTION QUOTIENT

### Collaboration has a score now.

CAVY doesn't just grade what you submit — it measures your CIQ: how well you question, edit, and reason alongside AI, session by session.

3  
Pillars measured

14  
Behavioral signals

### Create an account

Set up your CAVY login.

Student

Teacher

Full Name

Asha Patel

Email

asha@demo.edu

Password

.....

Enrolment No.  
(optional)

BSC-101

Create Account

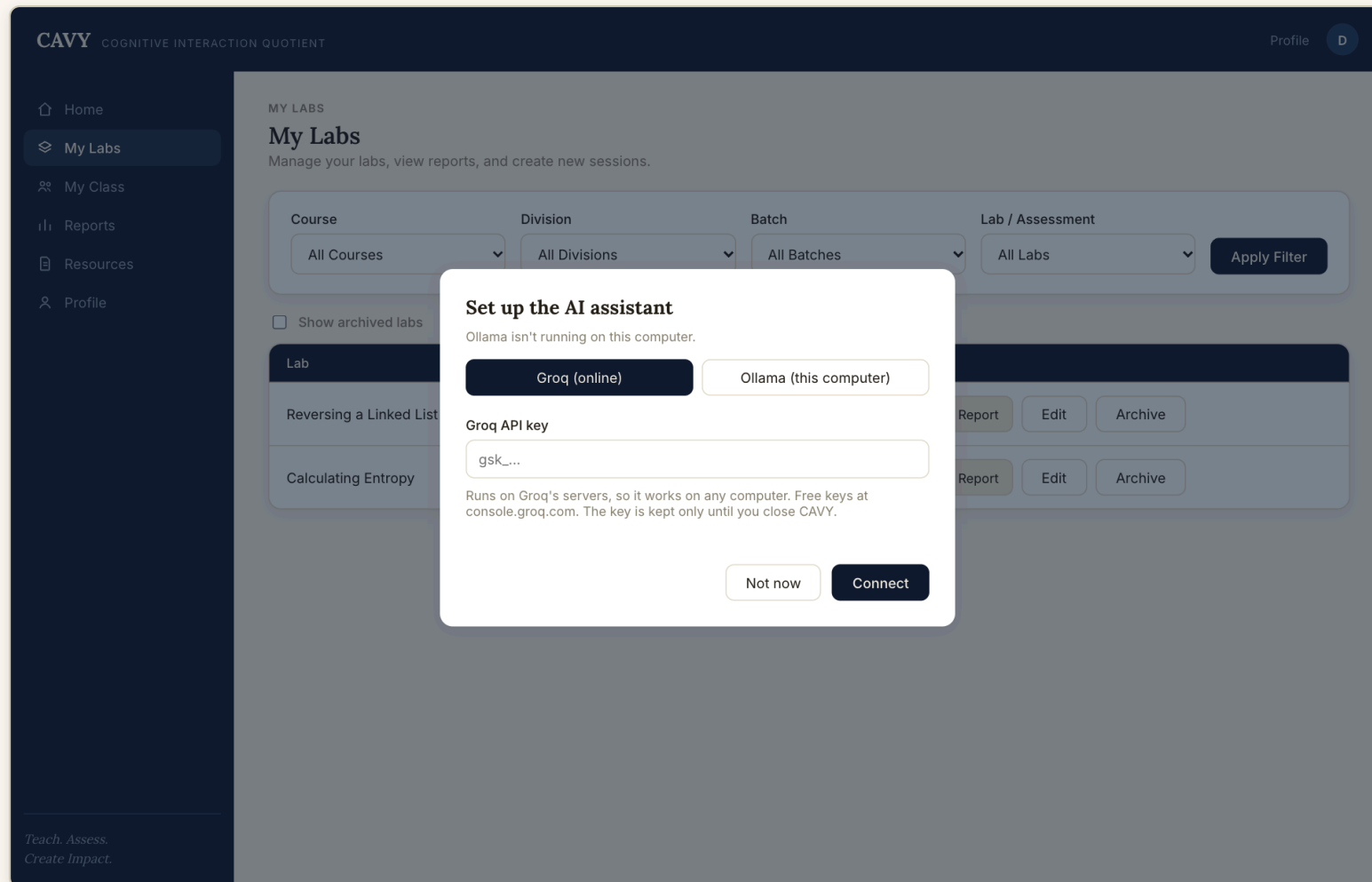
Already have an account? [Log in](#)

The sign-up form filled in, ready to submit.

## 2 • The teacher sets up a class

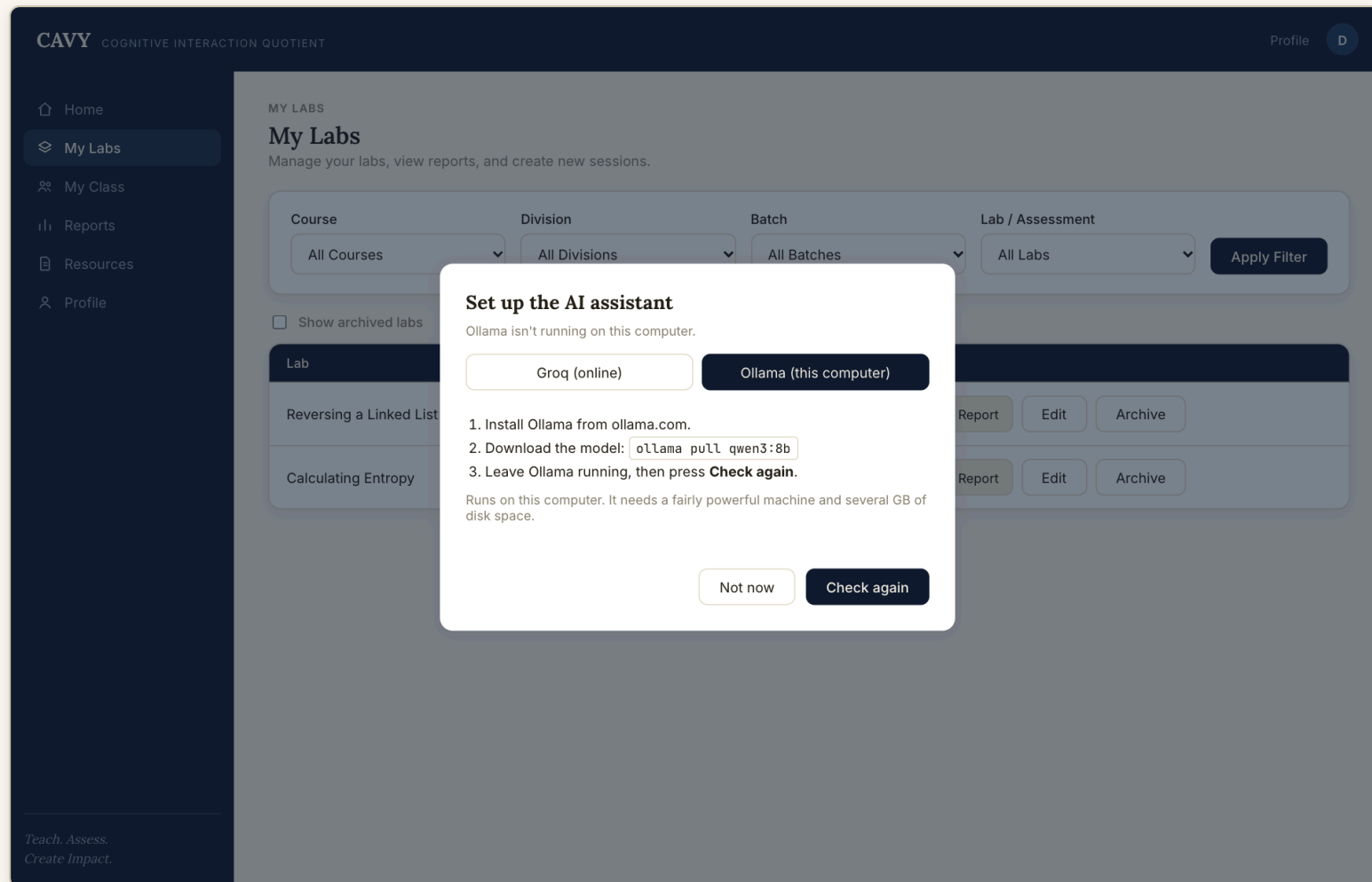
A teacher signs in, builds a class and publishes a lab.

## AI assistant setup (Groq)



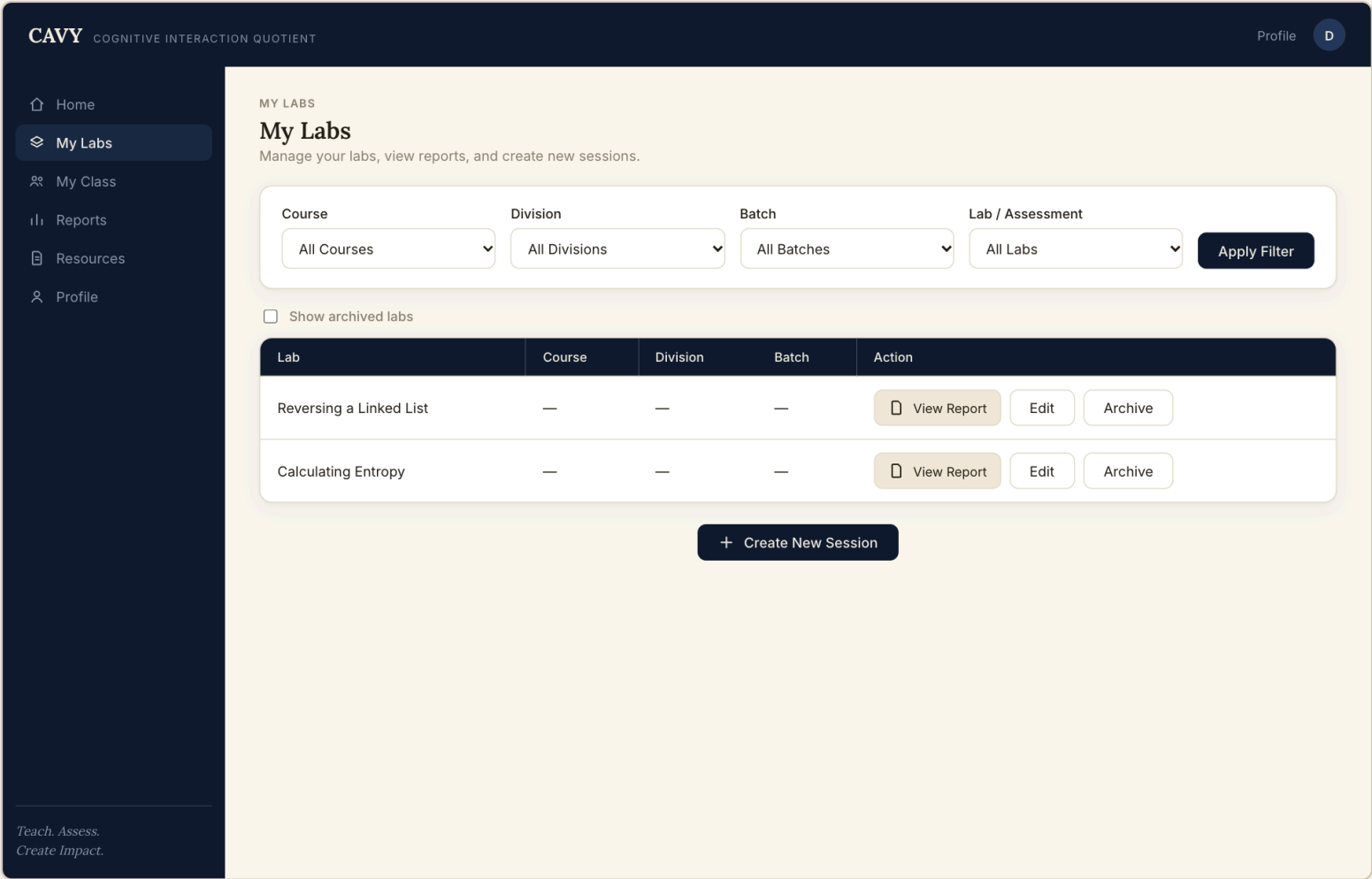
If the class AI assistant isn't available, the teacher is guided through setting it up: paste a Groq key (works from any computer), or use Ollama on the host computer.

## AI assistant setup (Ollama)



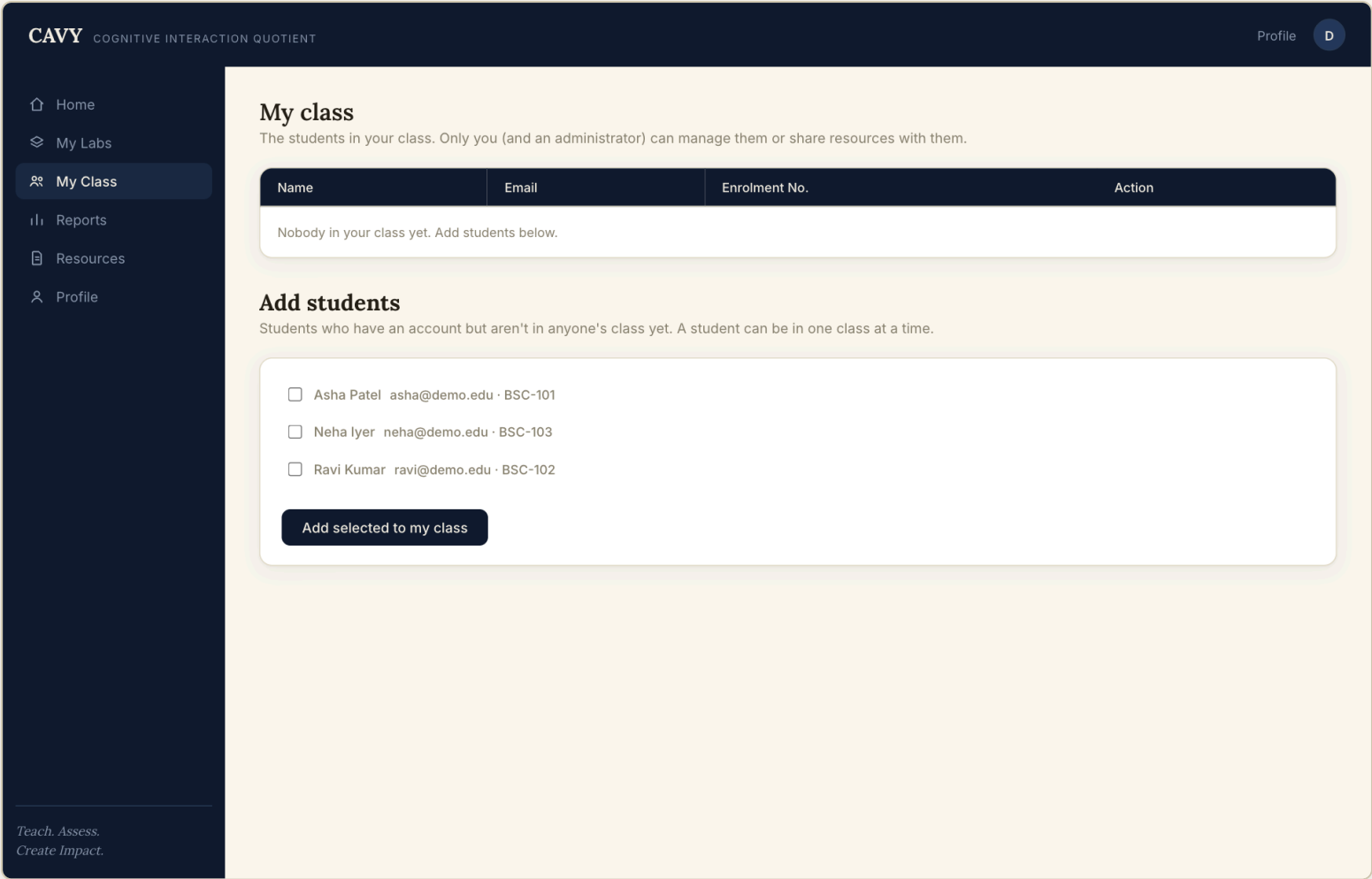
The Ollama option explains exactly what to install and checks whether it is ready.

# After signing in (My Labs)



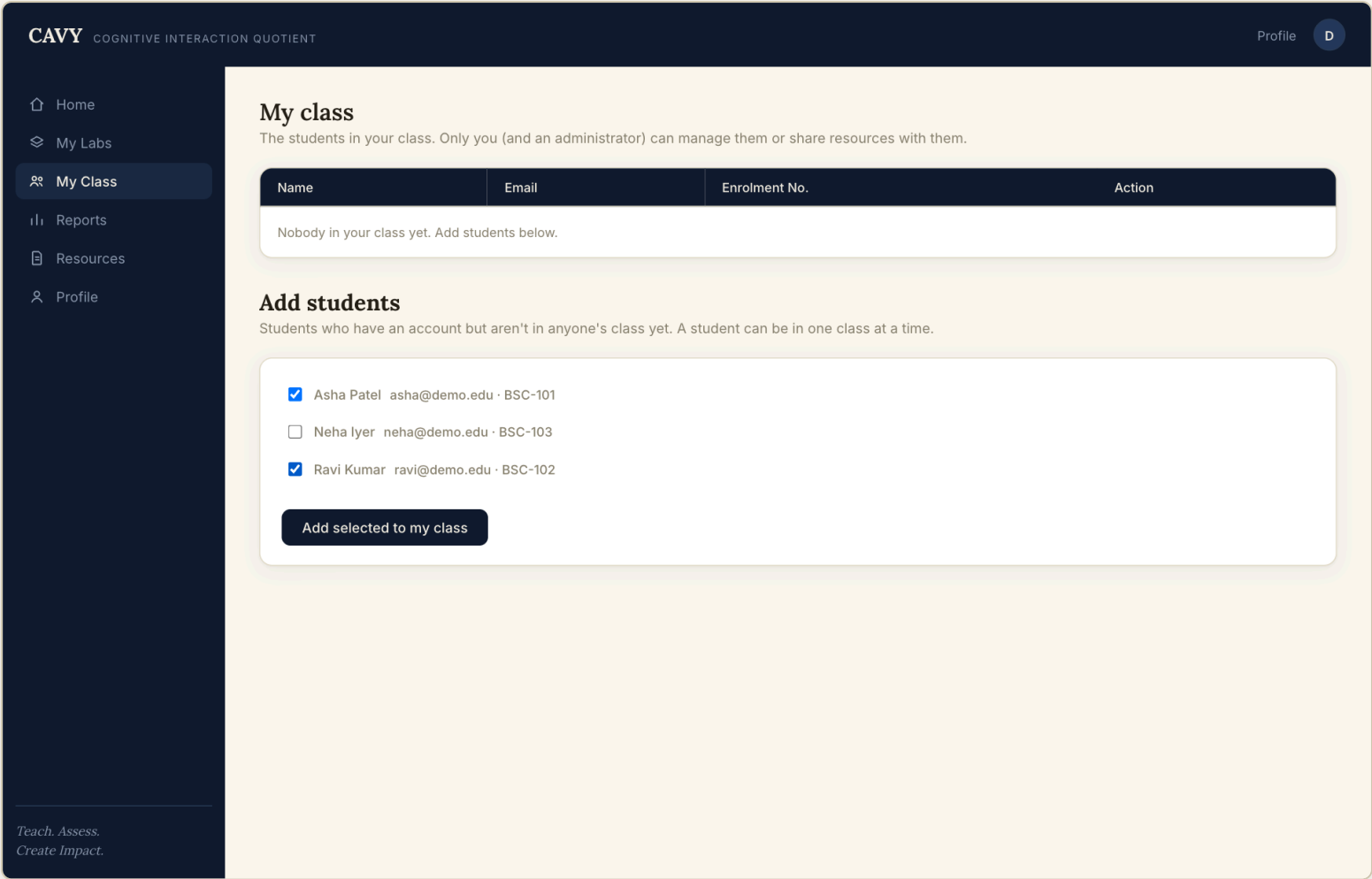
A teacher who signs in lands on My Labs. The sidebar's Home item opens the dashboard; the other items cover the class, reports and resources.

# My Class (before adding students)



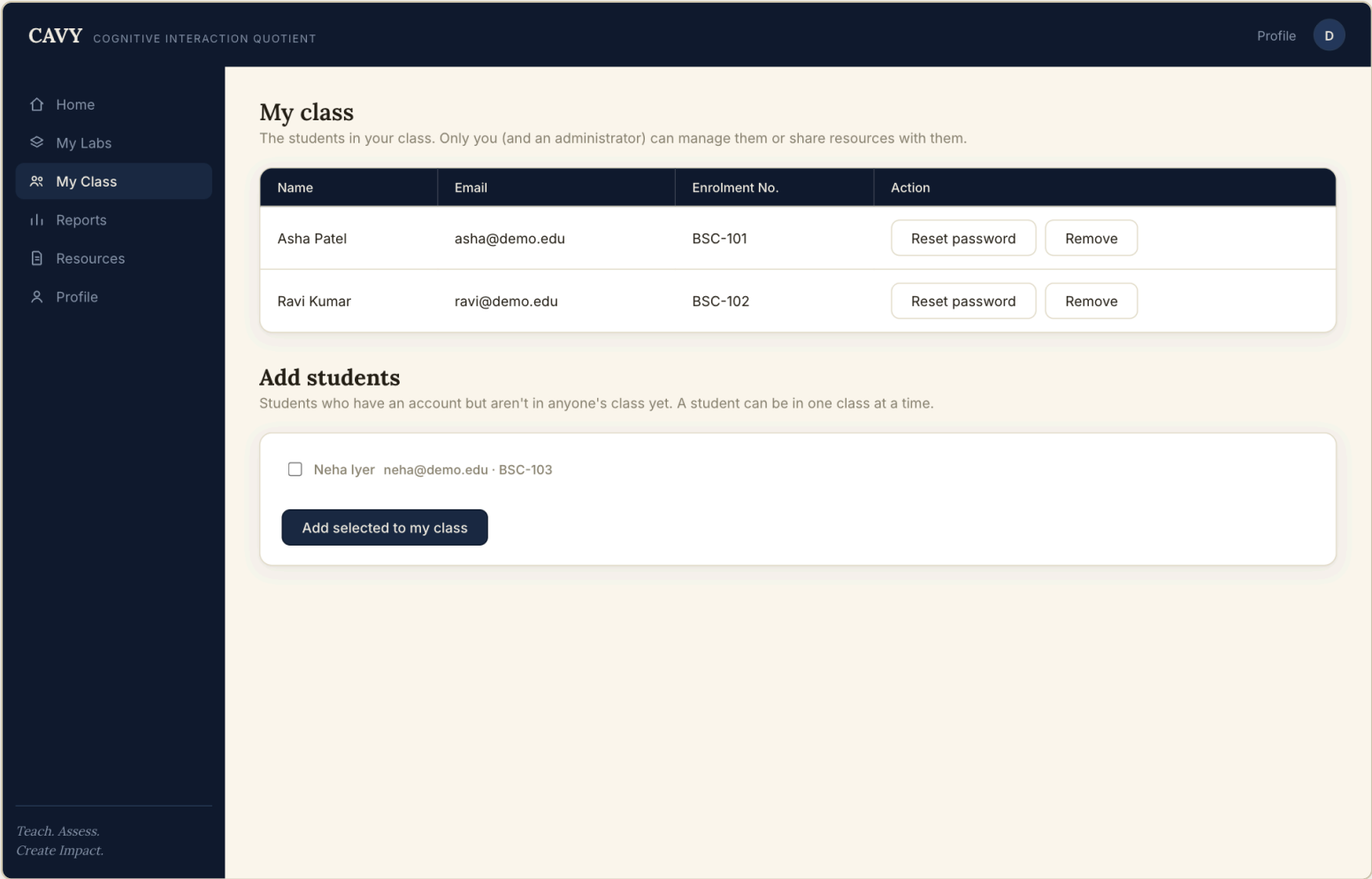
A teacher's class starts empty. Students who have an account but belong to nobody's class are listed below, ready to be added.

# My Class (choosing students to add)



Tick the students to bring into this class. A student can be in only one teacher's class.

# My Class (after adding)



Added students appear in the class table. From here the teacher can reset a student's password or remove them from the class.

# Create New Session (empty form)

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile

D

Home

My Labs

My Class

Reports

Resources

Profile

Back to Dashboard

Create New Session

Set up a new lab with learning, exploration, and assessment stages.

Session Details

Session Title

\*

Enter session title

Course

e.g. B.Sc. Data Science

Topic

Enter topic

Division

e.g. A

Batch

e.g. 2026

Difficulty

Medium

Description

Enter a brief description about this session...

Learning Workflow

Configure the duration and AI mode for each stage of the session.

Learning

Duration (minutes)

Exploration

Duration (minutes)

Assessment

Duration (minutes)

Teach. Assess.

Create Impact.

Creating a lab: title, course, cohort, topic, difficulty and description, then the duration and AI-assistance level for each of the three stages.

# Create New Session (filled in, top)

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile

D

Home

My Labs

My Class

Reports

Resources

Profile

Back to Dashboard

Create New Session

Set up a new lab with learning, exploration, and assessment stages.

Session Details

Session Title

Loops and Sums

Course

B.Sc. Computer Science

Topic

Iteration

Division

A

Batch

2026

Difficulty

Easy

Description

Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.

Learning Workflow

Configure the duration and AI mode for each stage of the session.

Learning

Duration (minutes)

Exploration

Duration (minutes)

Assessment

Duration (minutes)

Teach. Assess.

Create Impact.

The details of the new lab filled in.

## Create New Session (stage settings)

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile 

D

Home

My Labs

My Class

Reports

Resources

Profile

Teach. Assess.  
Create Impact.

Loops and Sums

Course

B.Sc. Computer Science

Topic

Iteration

Division

A

Batch

2026

Difficulty

Easy

Description

Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.

Learning Workflow

Configure the duration and AI mode for each stage of the session.

Learning

Duration (minutes)

20

AI Mode

Full Assistance

Exploration

Duration (minutes)

20

AI Mode

Full Assistance

Assessment

Duration (minutes)

20

AI Mode

Restricted Assistance

Create Lab →

Each stage has its own time limit and level of AI help: full assistance while learning, restricted for exploration, none for the assessment.

# My Labs (new lab published)

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile

D

Home

**My Labs**

My Class

Reports

Resources

Profile

MY LABS

### My Labs

Manage your labs, view reports, and create new sessions.

Course

Division

Batch

Lab / Assessment

All Courses

All Divisions

All Batches

All Labs

Apply Filter

☐ Show archived labs

| Lab                     | Course                 | Division | Batch | Action                                                    |
|-------------------------|------------------------|----------|-------|-----------------------------------------------------------|
| Reversing a Linked List | —                      | —        | —     | <div>View Report</div> <div>Edit</div> <div>Archive</div> |
| Calculating Entropy     | —                      | —        | —     | <div>View Report</div> <div>Edit</div> <div>Archive</div> |
| Loops and Sums          | B.Sc. Computer Science | A        | 2026  | <div>View Report</div> <div>Edit</div> <div>Archive</div> |

+ Create New Session

Teach. Assess.

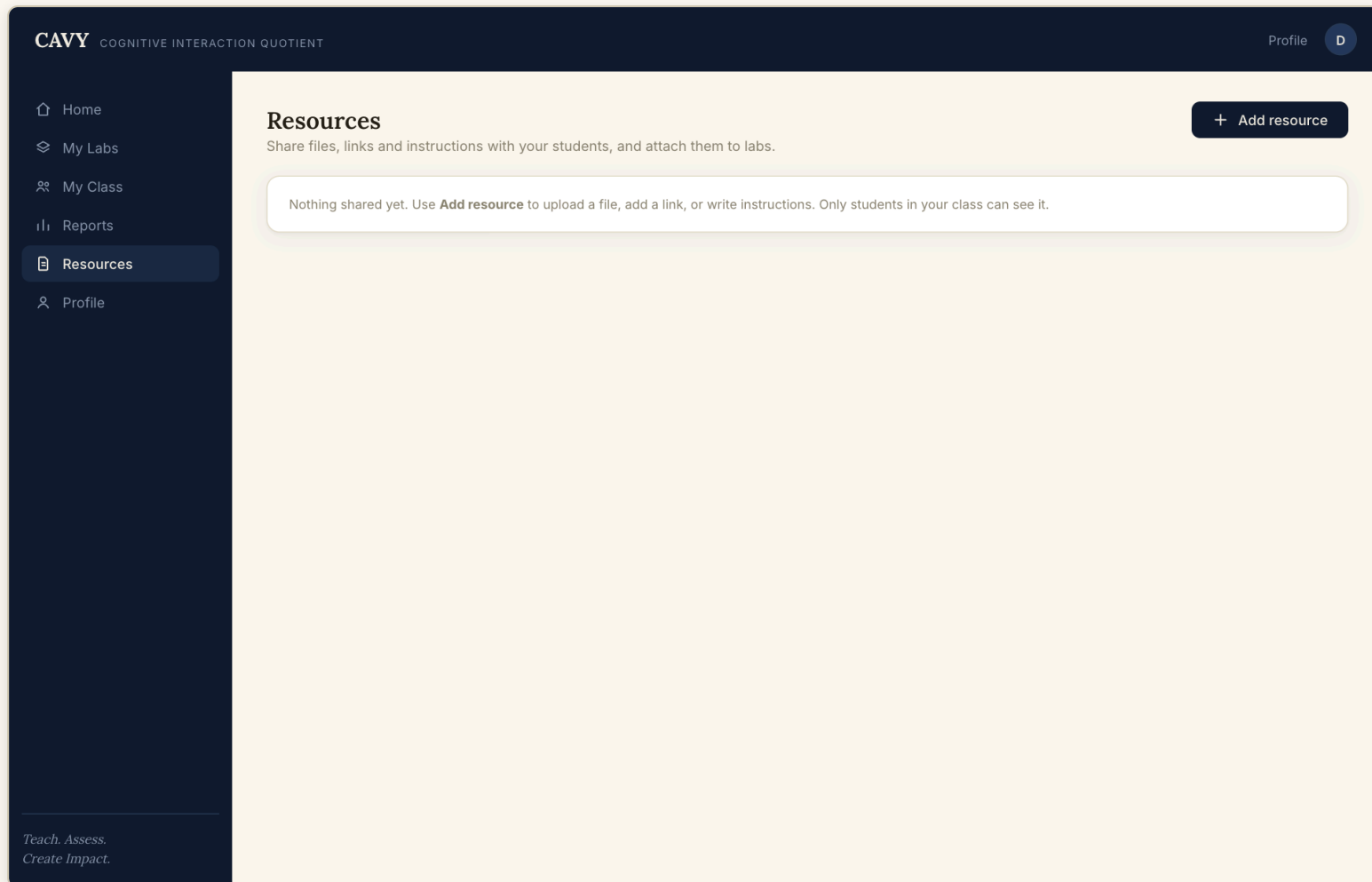
Create Impact.

The new lab is now listed. Students see it on their Home screen immediately.

## 3 • The teacher shares resources

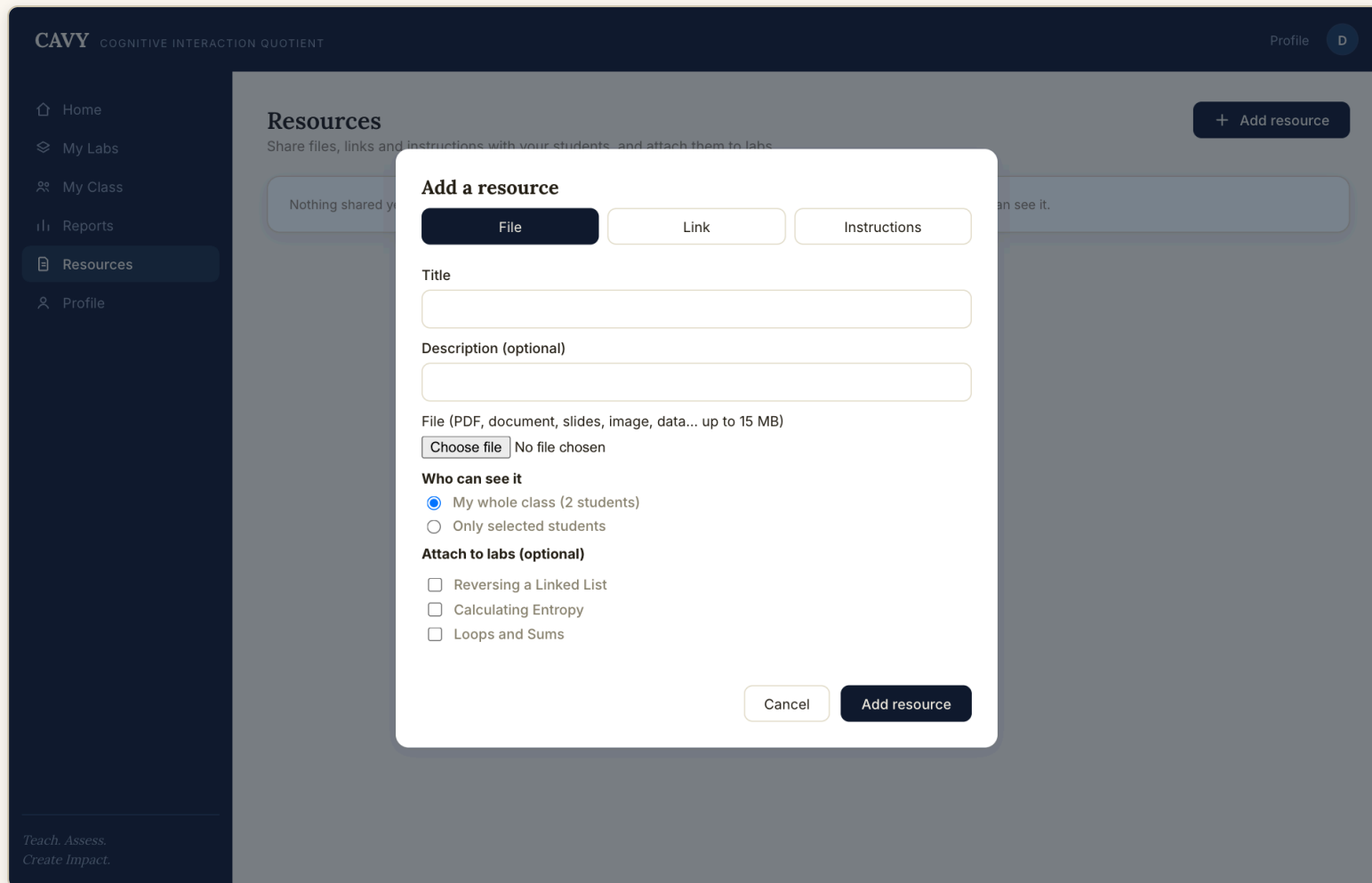
Files, links and written instructions, for the whole class or chosen students, attached to labs.

# Teacher Resources (empty)



Teachers share reference material with their class: files, links and written instructions.

## Add resource: File



Upload a PDF, document, slides or image (up to 15 MB). Programs and scripts are refused for safety.

## Add resource: File (filled in)

The screenshot shows the CAVY interface with a sidebar on the left containing links to Home, My Labs, My Class, Reports, Resources (highlighted), and Profile. The main content area is titled 'Resources' and has a '+ Add resource' button. A modal window titled 'Add a resource' is open, showing three tabs: 'File' (selected), 'Link', and 'Instructions'. The 'File' tab contains the following fields and options:

- Title:** A text input field containing 'Loops-cheat-sheet'.
- Description (optional):** A text input field containing 'A one-page summary of loop syntax'.
- File (PDF, document, slides, image, data... up to 15 MB):** A 'Choose file' button followed by the text 'Loops-cheat-sheet.pdf'.
- Who can see it:** Two radio button options: 'My whole class (2 students)' (selected) and 'Only selected students'.
- Attach to labs (optional):** Three checkbox options: 'Reversing a Linked List' (unchecked), 'Calculating Entropy' (unchecked), and 'Loops and Sums' (checked).

At the bottom of the modal are 'Cancel' and 'Add resource' buttons. The background of the interface is dark blue with white text, and the modal is white with dark blue text.

A file chosen, shared with the whole class, and attached to the 'Loops and Sums' lab.

## Add resource: Link

The screenshot shows the CAVY (Cognitive Interaction Quotient) interface. A modal dialog titled "Add a resource" is open, allowing a user to add a new resource. The dialog has three tabs: "File", "Link" (which is selected), and "Instructions".

The "Link" tab contains the following fields and options:

- Title:** A text input field containing "Python tutorial: for loops".
- Description (optional):** An empty text input field.
- Web address:** A text input field containing the URL "https://docs.python.org/3/tutorial/controlflow.html#for-statements".
- Who can see it:** Two radio button options: "My whole class (2 students)" (which is selected) and "Only selected students".
- Attach to labs (optional):** Three checkbox options: "Reversing a Linked List", "Calculating Entropy", and "Loops and Sums".

At the bottom of the dialog are two buttons: "Cancel" and "Add resource".

In the background, the "Resources" section of the interface is visible, showing a list of resources with columns for "File", "Loops", and "Sums". A resource titled "Loops-cheat-sheet" is visible, with a date of "03/10/2026" and a "Shared with you" status. There are "Open" and "Delete" buttons for this resource.

A web address. Students open it in their normal browser.

## Add resource: Instructions (chosen students)

The screenshot shows the CAVY (Cognitive Interaction Quotient) interface. A modal titled 'Add a resource' is open, with the 'Instructions' tab selected. The modal contains the following fields and options:

- Title:** A text input field containing 'Before you start'.
- Description (optional):** A text input field containing 'Read this first'.
- Instructions:** A text area containing a list of instructions:
  1. Use a for loop with range().
  2. Print each number on its own line.
  3. Keep a running total in a variable.
  4. Run your code before you submit.
- Who can see it:** Radio buttons for 'My whole class (2 students)' and 'Only selected students'. The 'Only selected students' option is selected.
- Attach to labs (optional):** Checkboxes for 'Reversing a Linked List', 'Calculating Entropy', and 'Loops and Sums'. The 'Loops and Sums' checkbox is checked.

At the bottom of the modal are 'Cancel' and 'Add resource' buttons. In the background, a 'Resources' list is visible, showing a link to a Python document with the title 'Python', a URL, a date '03/10/2026', and a button 'Open link'.

Written instructions shared with selected students only. A teacher can only choose from their own class.

## Teacher Resources (three items)

The screenshot displays the 'Resources' section of the CAVY platform. The interface includes a dark blue sidebar on the left with navigation links: Home, My Labs, My Class, Reports, Resources (highlighted), and Profile. The main content area has a header 'Resources' with a sub-header 'Share files, links and instructions with your students, and attach them to labs.' and a '+ Add resource' button. Three resource cards are shown:

- Instructions: Before you start**
  - Read this first
  - Loops and Sums
  - 03/10/2026
  - Shared with 1 selected student
  - Buttons: Read, Edit, Delete
- Link: Python tutorial: for loops**
  - https://docs.python.org/3/tutorial/controlflow.html#for-statements
  - 03/10/2026
  - Shared with your whole class
  - Buttons: Open link, Edit, Delete
- File: Loops-cheat-sheet**
  - A one-page summary of loop syntax
  - Loops-cheat-sheet.pdf · 17 KB
  - Loops and Sums
  - 03/10/2026
  - Shared with your whole class
  - Buttons: Open, Save as..., Edit, Delete

At the bottom of the sidebar, the text 'Teach. Assess. Create Impact.' is visible.

Everything the teacher has shared, with who can see it and which labs it is attached to. Each can be edited or deleted.

# Edit resource

The screenshot shows the CAVY Cognitive Interaction Quotient interface. A modal titled "Edit resource" is open, allowing a teacher to edit a resource. The modal contains the following fields and options:

- Title:** A text input field containing "Before you start".
- Description (optional):** A text input field containing "Read this first".
- Instructions:** A text area containing a list of instructions:
  1. Use a for loop with range().
  2. Print each number on its own line.
  3. Keep a running total in a variable.
  4. Run your code before you submit.
- Who can see it:** Radio buttons for "My whole class (2 students)" and "Only selected students". The "Only selected students" option is selected. Below are checkboxes for individual users: "Asha Patel asha@demo.edu" (checked) and "Ravi Kumar ravi@demo.edu" (unchecked).
- Attach to labs (optional):** Checkboxes for "Reversing a Linked List" (unchecked), "Calculating Entropy" (unchecked), and "Loops and Sums" (checked).

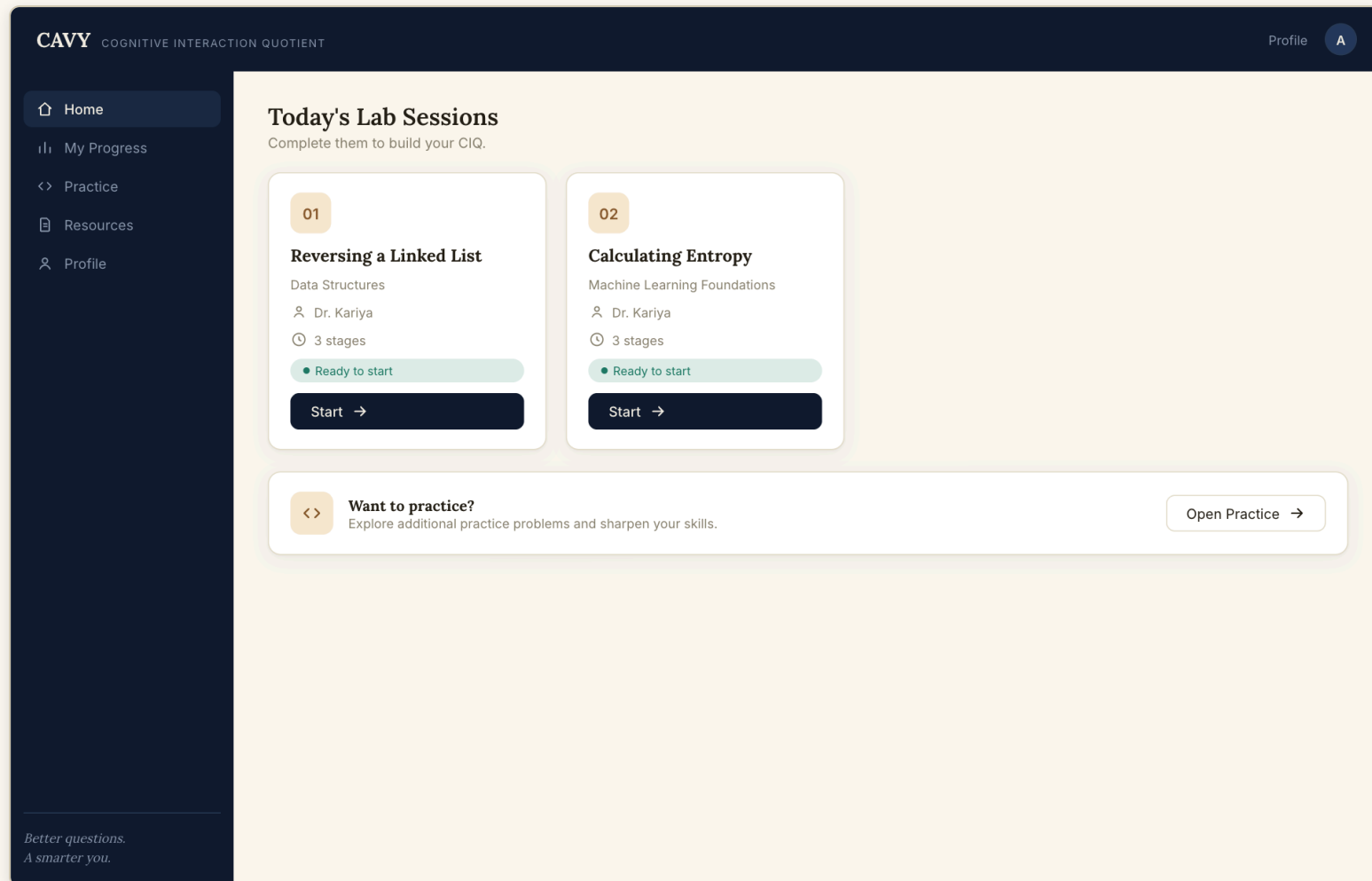
At the bottom of the modal are "Cancel" and "Save changes" buttons. The background interface shows a sidebar with navigation links (Home, My Labs, My Class, Reports, Resources, Profile) and a main area displaying a resource card for "Loops-cheat-sheet" with details like "A one-page summary of loop syntax", "Loops-cheat-sheet.pdf · 17 KB", and "03/10/2026".

Editing keeps the resource's type. A new file can replace the old one.

## 4 • The student starts a lab

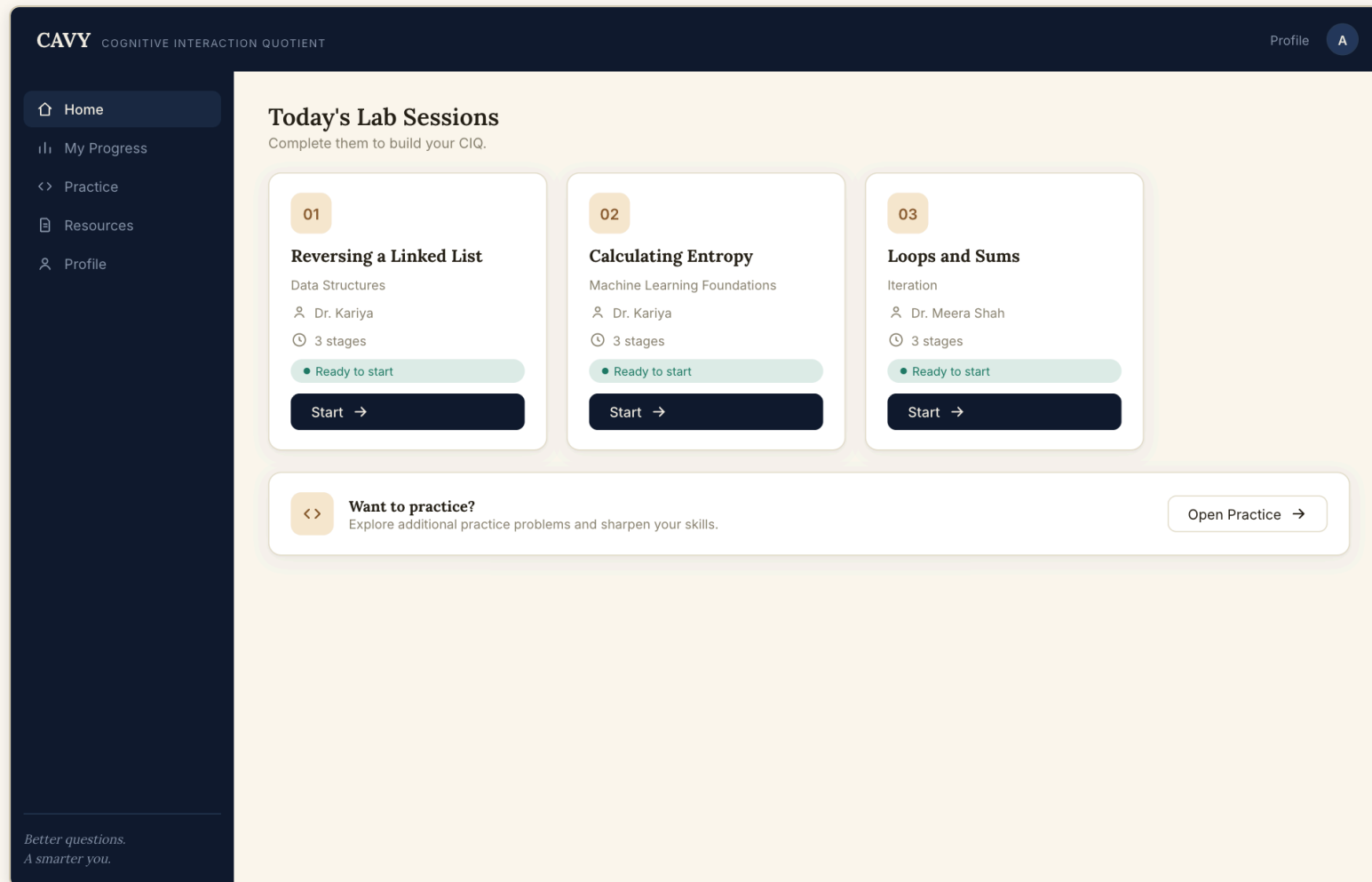
What a student sees when they sign in and open a lab.

# Student Home



A new student lands on Home. Nothing has been published yet, so the list of lab sessions is empty apart from the demo labs.

## Student Home (lab sessions)



The student's Home: the lab sessions their teacher has published. New labs appear here by themselves, without refreshing.

# Lab stages

The screenshot displays the CAVY Cognitive Interaction Quotient interface. The top navigation bar includes the CAVY logo, the text 'COGNITIVE INTERACTION QUOTIENT', a 'Profile' link, and a user icon labeled 'A'. A dark sidebar on the left contains navigation links: 'Home', 'My Progress', 'Practice', 'Resources', and 'Profile'. The main content area is titled 'Loops and Sums' and features a progress indicator with three stages: 1. Learning, 2. Exploration, and 3. Assessment. Below the progress indicator, three stage cards are listed. Stage 1: Learning is active, showing 'AI Assistance: Full assistance' and a 'Ready to Start' status, with a 'Start' button. Stage 2: Exploration is locked, showing 'AI Assistance: Full assistance' and a 'Locked' status, with a 'Locked' button. Stage 3: Assessment is locked, showing 'AI Assistance: Restricted assistance' and a 'Locked' status, with a 'Locked' button. Below the stage cards, a section titled 'Resources for this lab' contains two resource cards. The first card, 'Instructions Before you start', includes the text 'Read this first', 'Loops and Sums', and 'From Dr. Meera Shah · 03/10/2026', with a 'Read' button. The second card, 'File Loops-cheat-sheet', includes the text 'A one-page summary of loop syntax', 'Loops-cheat-sheet.pdf · 17 KB', 'Loops and Sums', and 'From Dr. Meera Shah · 03/10/2026', with 'Open' and 'Save as...' buttons. The bottom of the sidebar features the text 'Better questions. A smarter you.'

CAVY COGNITIVE INTERACTION QUOTIENT Profile A

Home My Progress Practice Resources Profile

< Back to Dashboard

## Loops and Sums

1 Learning 2 Exploration 3 Assessment

1 Stage 1: Learning AI Assistance: Full assistance Ready to Start Start →

2 Stage 2: Exploration AI Assistance: Full assistance Locked Locked →

3 Stage 3: Assessment AI Assistance: Restricted assistance Locked Locked →

### Resources for this lab

Instructions Before you start

Read this first

Loops and Sums

From Dr. Meera Shah · 03/10/2026

Read

File Loops-cheat-sheet

A one-page summary of loop syntax

Loops-cheat-sheet.pdf · 17 KB

Loops and Sums

From Dr. Meera Shah · 03/10/2026

Open Save as...

Better questions. A smarter you.

Each lab has three stages: Learning, Exploration and Assessment, each with its own level of AI help. Later stages unlock as earlier ones are submitted.

## Lab stages (resources for this lab)

The screenshot displays the CAVY Cognitive Interaction Quotient interface. On the left is a dark blue sidebar with navigation links: Home, My Progress, Practice, Resources, and Profile. The main content area has a dark blue header with the CAVY logo and a user profile icon labeled 'A'. Below the header, a breadcrumb link 'Back to Dashboard' is visible. The main title 'Loops and Sums' is followed by a progress bar with three stages: 1 (Learning), 2 (Exploration), and 3 (Assessment). Stage 1 is active, showing 'Stage 1: Learning' with 'AI Assistance: Full assistance' and a 'Ready to Start' status. Stage 2 is 'Locked', and Stage 3 is 'Locked'. Below the stages, the 'Resources for this lab' section contains two cards. The first card, 'Instructions Before you start', includes a 'Read' button. The second card, 'File Loops-cheat-sheet', includes 'Open' and 'Save as...' buttons. The interface also features a footer with the text 'Better questions. A smarter you.'

CAVY COGNITIVE INTERACTION QUOTIENT

Profile A

< Back to Dashboard

### Loops and Sums

1 Learning 2 Exploration 3 Assessment

1 **Stage 1: Learning**

AI Assistance: Full assistance • Ready to Start

Start →

2 **Stage 2: Exploration**

AI Assistance: Full assistance Locked

Locked →

3 **Stage 3: Assessment**

AI Assistance: Restricted assistance Locked

Locked →

### Resources for this lab

**Instructions Before you start**

Read this first

Loops and Sums

From Dr. Meera Shah · 03/10/2026

Read

**File Loops-cheat-sheet**

A one-page summary of loop syntax

Loops-cheat-sheet.pdf · 17 KB

Loops and Sums

From Dr. Meera Shah · 03/10/2026

Open Save as...

Better questions.  
A smarter you.

The teacher's resources for this lab, shown right where the student starts working.

# 5 • The coding workspace

Writing and running code, asking the AI assistant, and the restricted Assessment stage.

# Workspace (start of the Learning stage)

The screenshot displays the CAVY workspace interface. At the top, the header shows 'CAVY COGNITIVE INTERACTION QUOTIENT' on the left, 'TIME REMAINING 19:58' and an 'End Session' button in the center, and a 'Profile' section with a blue circle containing the letter 'A' on the right.

The main workspace area is divided into three sections:

- Task Description:** Titled 'Loops and Sums — Learning', it includes tags for 'Learning', 'Full assistance', 'Difficulty: Easy', 'Est. Time: 20m', and a 'Hide Description' button. The task instruction reads: 'Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.'
- Code Editor:** A dark-themed editor with a file tab 'main.py'. It contains a single line of code: '1'. Above the editor are buttons for 'Run Code', 'Reset', 'Save', and 'Submit'. The status bar at the bottom indicates 'Python', 'Ln —, Col —', 'Spaces: 4', and 'Not saved yet'.
- AI Assistant:** A panel on the right titled '+ AI Assistant' with an information icon. It shows 'Assistant ready' and tabs for 'Chat' and 'Examples'. A list of capabilities includes: 'Understanding the problem', 'Explaining errors', 'Suggesting approaches', and 'Code examples'. Below this is a message: 'Try solving the task on your own first. Use me when you're stuck or want to verify your approach.' There are three example prompts: 'How do I get started on this?', 'Can you explain the error I'm seeing?', and 'Is my approach on the right track?'. At the bottom is an input field 'Ask anything...' with a send button and a disclaimer: 'AI can make mistakes. Verify important information.'

The left sidebar contains navigation links: 'Back to Lab', 'Loops and Sums', 'STAGES' (with 'Learning' selected), 'Exploration', 'Assessment', 'RESOURCES' (with 'Task Description' and 'Reference Notes'), and the footer text 'Learn. Think. Collaborate. Grow.'

The workspace: the task on top, a code editor, the output panel, and the AI assistant on the right. The countdown and the stage's AI level show in the header.

# Workspace (writing code)

The screenshot displays the CAVY workspace interface. At the top, the header includes the CAVY logo, the text "COGNITIVE INTERACTION QUOTIENT", a timer showing "TIME REMAINING 19:56", an "End Session" button, and a profile icon labeled "A".

On the left sidebar, there is a "Back to Lab" link, a section titled "Loops and Sums", and a "STAGES" list with "Learning" selected. Below this, there are "Exploration" and "Assessment" options. A "RESOURCES" section lists "Task Description" and "Reference Notes". At the bottom of the sidebar, the text "Learn. Think. Collaborate. Grow." is visible.

The main workspace area is titled "Loops and Sums — Learning". It includes tags for "Learning", "Full assistance", "Difficulty: Easy", "Est. Time: 20m", and a "Hide Description" button. Below these is a text box with the instruction: "Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum."

In the center is a code editor for a file named "main.py". The code is as follows:

```
1 total = 0
2 for i in range(1, 6): total += i; print(i)
3 print("Sum:", total)
4
```

Above the code editor are buttons for "Run Code", "Reset", "Save", and "Submit". The status bar at the bottom of the editor shows "Python Ln 4, Col 1 Spaces: 4" and "Autosaved just now".

On the right side, there is an "AI Assistant" panel. It shows "Assistant ready" and tabs for "Chat" and "Examples". The chat area contains the message: "Hi! I can help you with:" followed by a list of capabilities: "Understanding the problem", "Explaining errors", "Suggesting approaches", and "Code examples". Below this is a prompt: "Try solving the task on your own first. Use me when you're stuck or want to verify your approach." There are also input boxes for "How do I get started on this?", "Can you explain the error I'm seeing?", and "Is my approach on the right track?". At the bottom of the chat is an "Ask anything..." input field and a "Send" button. A note at the very bottom of the panel states: "AI can make mistakes. Verify important information."

As the student types, snapshots of their work are recorded as learning evidence (never raw keystrokes).

# Workspace (running code)

The screenshot displays the CAVY workspace interface. At the top, the header shows 'CAVY COGNITIVE INTERACTION QUOTIENT' on the left, 'TIME REMAINING 19:55' and 'End Session' on the right, and a 'Profile' button with a user icon 'A'.

The left sidebar contains navigation links: 'Back to Lab', 'Loops and Sums' (selected), 'STAGES' (Learning, Exploration, Assessment), and 'RESOURCES' (Task Description, Reference Notes). At the bottom of the sidebar is the text 'Learn. Think. Collaborate. Grow.'

The main workspace area is titled 'Loops and Sums — Learning'. It includes tags for 'Learning', 'Full assistance', 'Difficulty: Easy', 'Est. Time: 20m', and a 'Hide Description' button. Below this is a text box with the instruction: 'Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.'

The code editor shows a file named 'main.py' with the following Python code:

```
1 total = 0
2 for i in range(1, 6): total += i; print(i)
3 print("Sum:", total)
4
```

Buttons for 'Run Code', 'Reset', 'Save', and 'Submit' are located above the editor. The status bar at the bottom of the editor indicates 'Python Ln 4, Col 1 Spaces: 4' and 'Autosaved just now'.

Below the editor is the 'Output: main.py' section, which displays the program's output:

```
1
2
3
4
5
Sum: 15
```

On the right side of the workspace is the 'AI Assistant' panel. It shows 'Assistant ready' and tabs for 'Chat' and 'Examples'. A list of capabilities is provided: 'Understanding the problem', 'Explaining errors', 'Suggesting approaches', and 'Code examples'. A message encourages the user to try solving the task on their own first. Below this are several example prompts: 'How do I get started on this?', 'Can you explain the error I'm seeing?', and 'Is my approach on the right track?'. At the bottom is an input field 'Ask anything...' with a send button and a disclaimer: 'AI can make mistakes. Verify important information.'

Code runs safely on the student's own computer with a time limit. The output appears below the editor.

# AI assistant (question typed)

The screenshot displays the CAVY Cognitive Interaction Quotient interface. The top header shows the session title "CAVY COGNITIVE INTERACTION QUOTIENT", a timer at 19:54, and buttons for "End Session" and "Profile". The left sidebar contains navigation links: "Back to Lab", "Loops and Sums" (selected), "STAGES" (Learning, Exploration, Assessment), and "RESOURCES" (Task Description, Reference Notes). The main content area is titled "Loops and Sums — Learning" and includes buttons for "Learning", "Full assistance", "Difficulty: Easy", "Est. Time: 20m", and "Hide Description". A task description box states: "Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum." Below this is a code editor for "main.py" with the following code:

```
1 total = 0
2 for i in range(1, 6): total += i; print(i)
3 print("Sum:", total)
4
```

The editor has buttons for "Run Code", "Reset", "Save", and "Submit". The status bar indicates "Python Ln 4, Col 1 Spaces: 4" and "Autosaved just now". Below the editor is the "Output: main.py" window showing the execution results:

```
1
2
3
4
5
Sum: 15
```

On the right, the "AI Assistant" panel is active, showing "Assistant ready" and tabs for "Chat" and "Examples". It lists capabilities: "Understanding the problem", "Explaining errors", "Suggesting approaches", and "Code examples". A message encourages the user to try solving the task first. Below are several question prompts: "How do I get started on this?", "Can you explain the error I'm seeing?", "Is my approach on the right track?", and a typed question "does my loop print 1 to 5 but not 6?". A disclaimer at the bottom states: "AI can make mistakes. Verify important information."

The assistant knows the task, the student's current code and the last run's output.

# AI assistant (answer)

The screenshot displays the CAVY Cognitive Interaction Quotient interface. The top header shows the CAVY logo, session time (19:28), and an 'End Session' button. The left sidebar contains navigation links for 'Back to Lab', 'Loops and Sums' (selected), and 'STAGES' (Learning, Exploration, Assessment). Below this are 'RESOURCES' (Task Description, Reference Notes) and a footer with the text 'Learn. Think. Collaborate. Grow.'

The main workspace is titled 'Loops and Sums — Learning'. It includes buttons for 'Learning', 'Full assistance', 'Difficulty: Easy', 'Est. Time: 20m', and 'Hide Description'. A task instruction box states: 'Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.'

The code editor shows a Python file named 'main.py' with the following code:

```
1 total = 0
2 for i in range(1, 6): total += i; print(i)
3 print("Sum:", total)
4
```

Buttons for 'Run Code', 'Reset', 'Save', and 'Submit' are visible above the editor. The output area below the editor shows the execution results:

```
Output: main.py
1
2
3
4
5
Sum: 15
```

On the right, the 'AI Assistant' panel is active, showing a chat interface. The assistant's response to the user's question 'Why does my loop print 1 to 5 but not 6?' is as follows:

**You:** Why does my loop print 1 to 5 but not 6?

**Assistant:**

The loop uses `range(1, 6)`, which in Python generates numbers **starting from 1 up to but not including 6**. This is why the loop prints `1` through `5`—the range stops at `5`. If you wanted to include `6`, you'd need to adjust the range to `range(1, 7)` (since the end value is exclusive).

Your code is correct as written, and the sum `15` is accurate for numbers `1` to `5`.

The chat interface includes a text input field 'Ask anything...' and a confirmation message: 'AI can make mistakes. Verify important information.'

The assistant's reply. Every question and answer is recorded for the learning analysis.

## Assessment stage (AI restricted)

CAVY COGNITIVE INTERACTION QUOTIENT

TIME REMAINING 19:58 End Session Profile A

### Loops and Sums — Assessment

Assessment + Restricted assistance Difficulty: Easy Est. Time: 20m Hide Description

Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.

main.py + Run Code Reset Save Submit →

```
1
```

Python Ln —, Col — Spaces: 4 Not saved yet

#### + AI Assistant

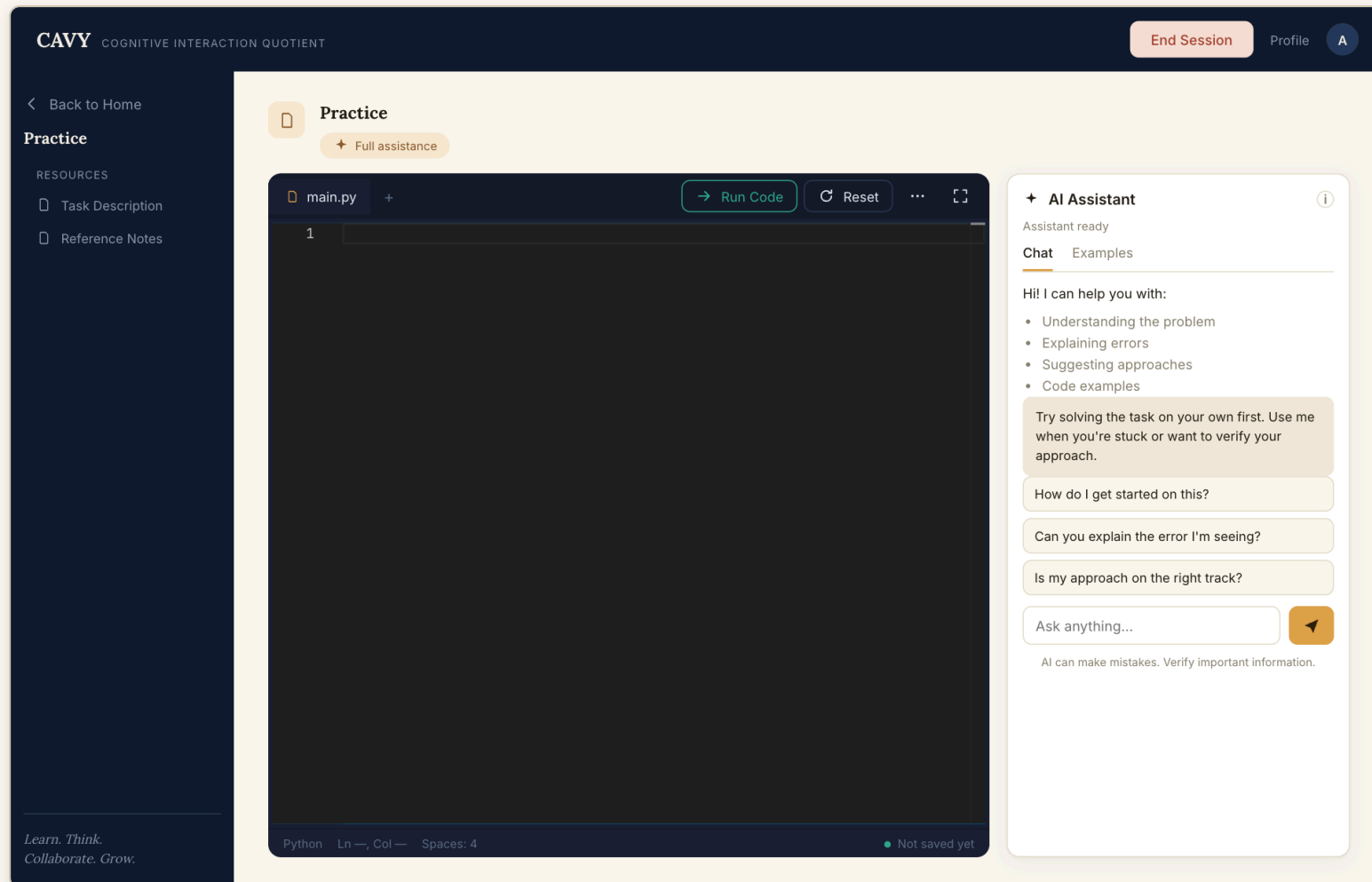
AI assistance is restricted during this stage

Ask anything...

Learn. Think. Collaborate. Grow.

In the Assessment stage the AI assistant is restricted, and the server enforces it as well. This is what makes the independent-work evidence meaningful.

# Practice workspace

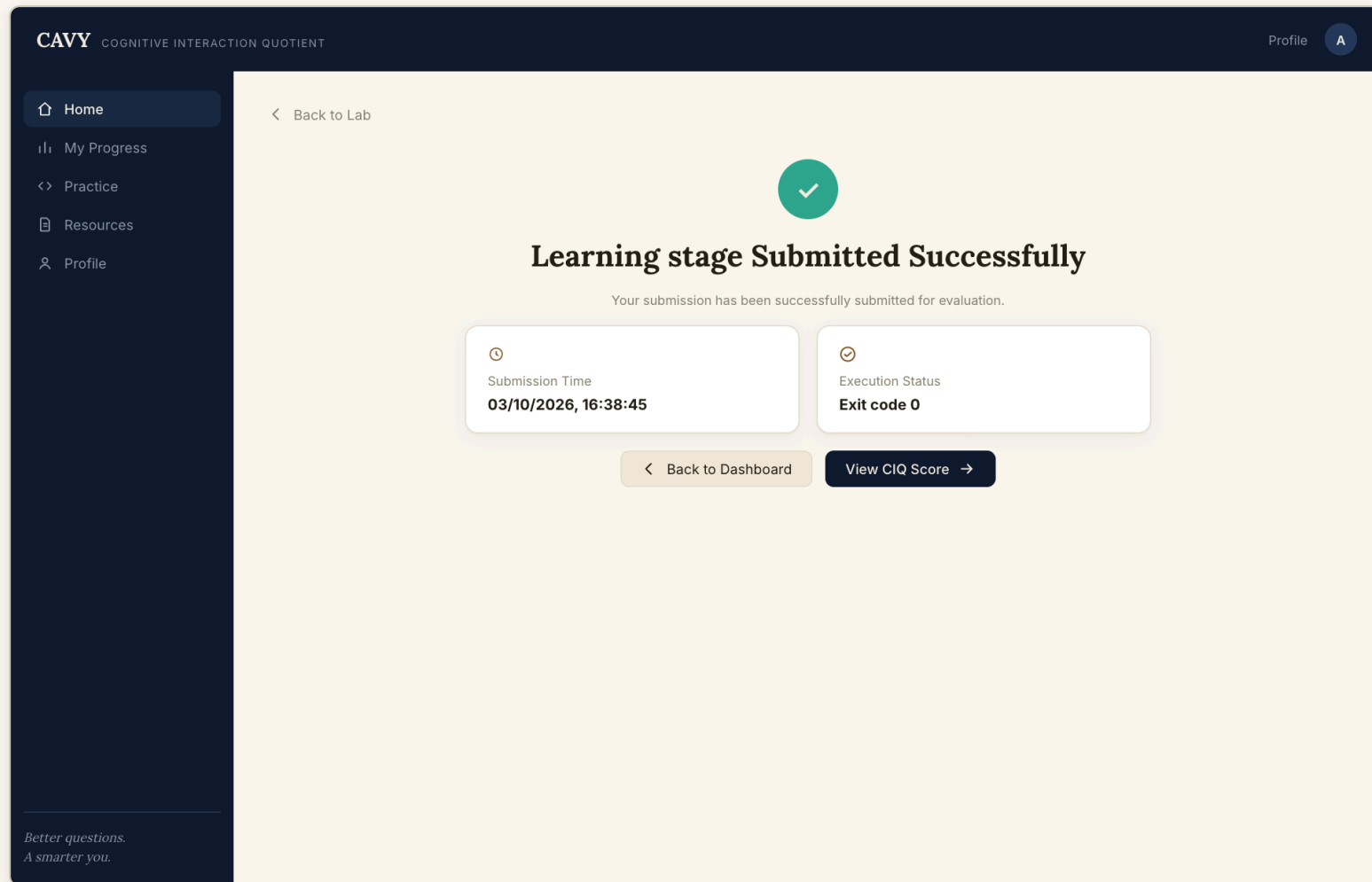


Practice is a free-form workspace that isn't part of any lab. It uses the same editor, runner and assistant, and is also recorded.

# 6 • Submitting and scores

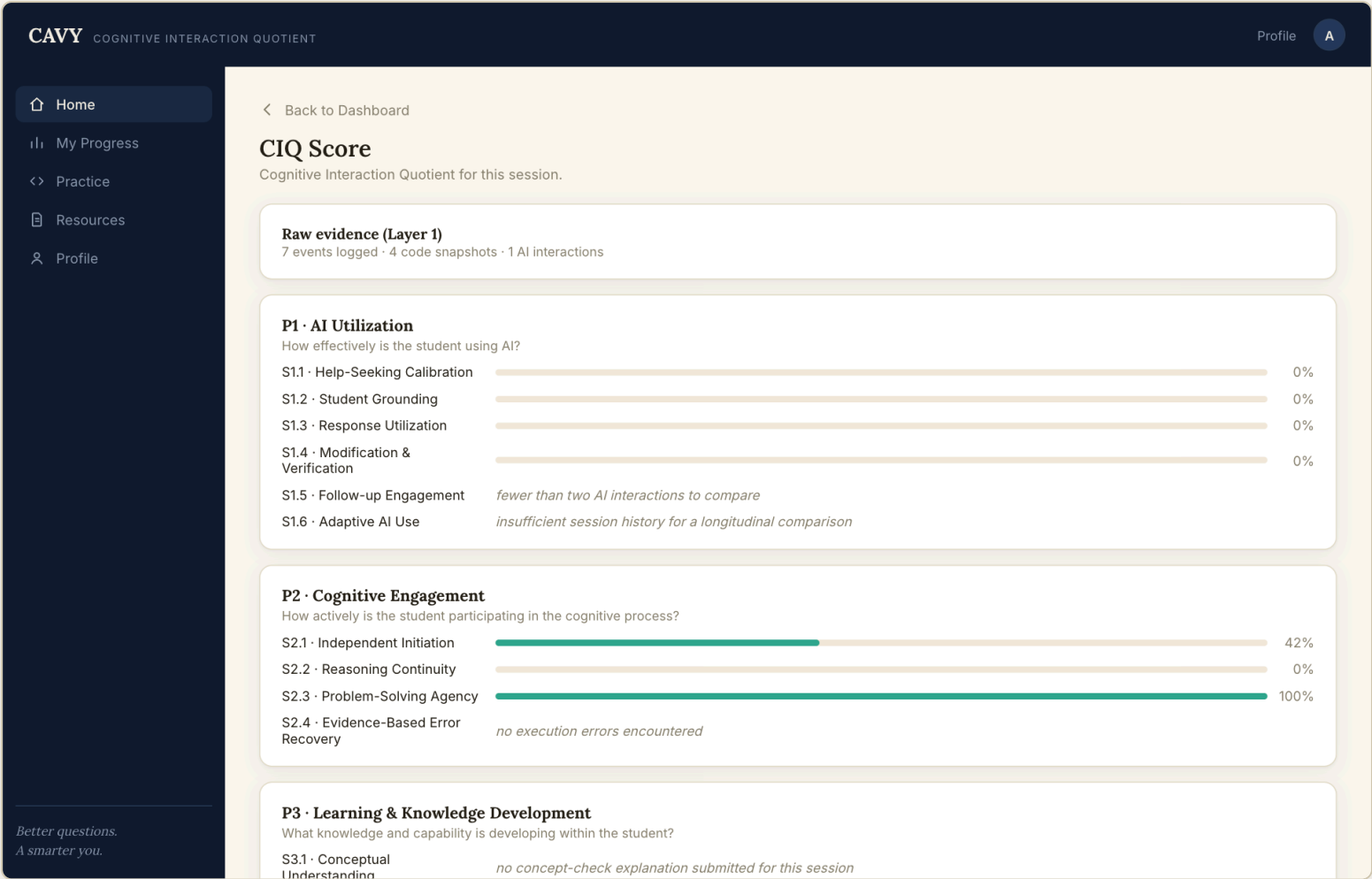
Submitting work, explaining a solution, and the Cognitive Interaction Quotient (CIQ) score.

# Submission confirmation



Confirmation that the work was submitted. The teacher's dashboard updates at the same moment. The student can open the score straight away.

# CIQ score (Learning stage)



The Cognitive Interaction Quotient for this session: three pillars and fourteen signals, from how the student asked, edited and checked their work. A signal without enough evidence says so instead of guessing.

## Concept check (dialog)

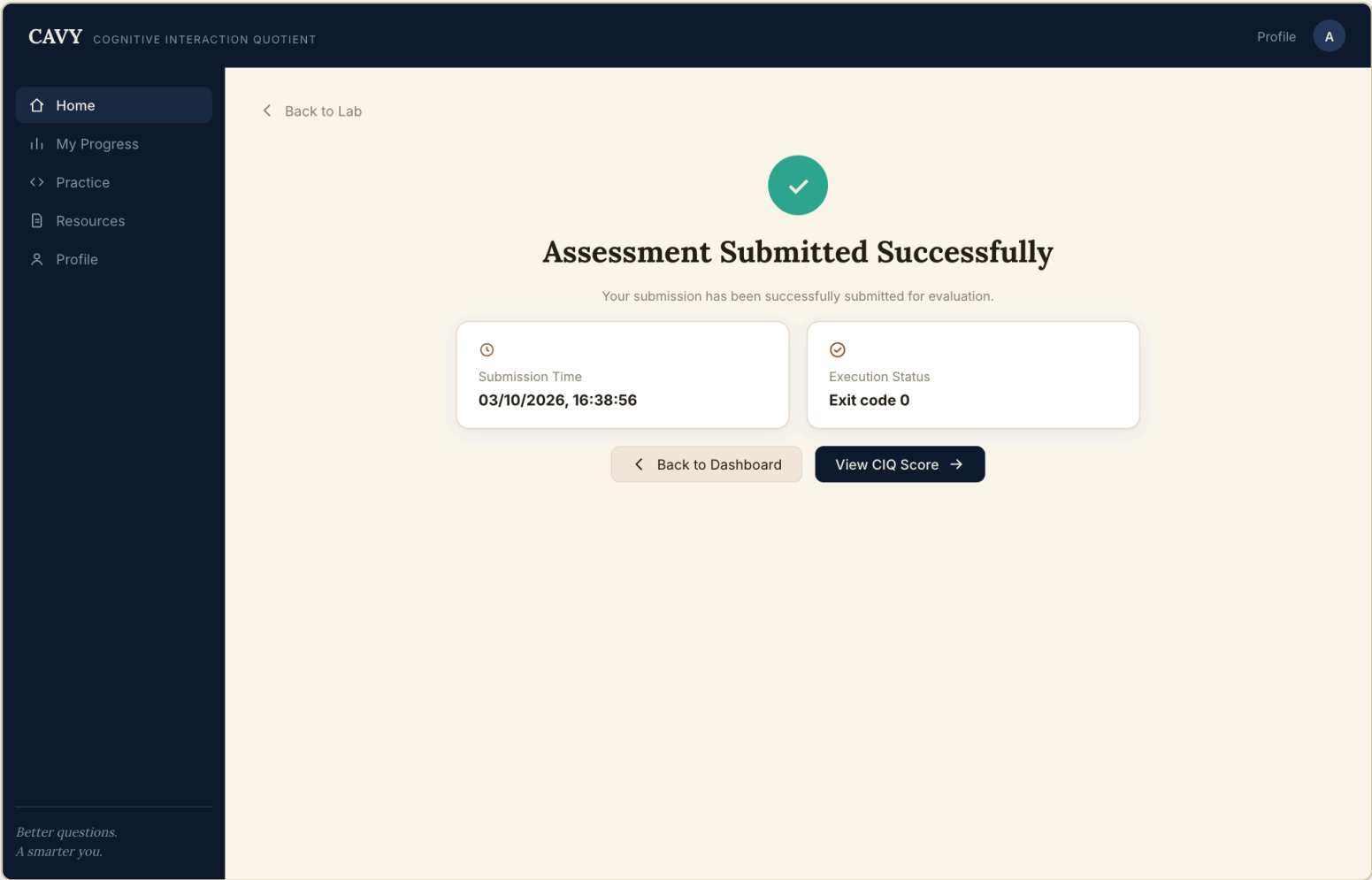
The screenshot shows the CAVY Cognitive Interaction Quotient interface. The top bar includes the CAVY logo, a session timer at 19:55, an 'End Session' button, and a profile icon. The left sidebar contains navigation links: 'Back to Lab', 'Loops and Sums' (selected), 'STAGES' (Learning, Exploration, Assessment), and 'RESOURCES' (Task Description, Reference Notes). The main area is titled 'Loops and Sums — Assessment' and includes buttons for 'Assessment', 'Restricted assistance', 'Difficulty: Easy', 'Est. Time: 20m', and 'Hide Description'. The task instruction is: 'Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.' Below this is a code editor for 'main.py' with the following code:

```
1 total = 0
2 for i in range(1, 6):
3     print(i)
4     total += i
```

The code has been executed, and the output is shown at the bottom: '1', '2', '3', '4', '5', and 'Sum: 15'. A modal dialog titled 'Before you submit' is displayed in the center, asking the user to explain their solution. The dialog includes a text input field and two buttons: 'Skip' and 'Submit Explanation'. On the right side of the interface, there is an 'AI Assistant' panel with a note that 'AI assistance is restricted during this stage' and a text input field for asking questions.

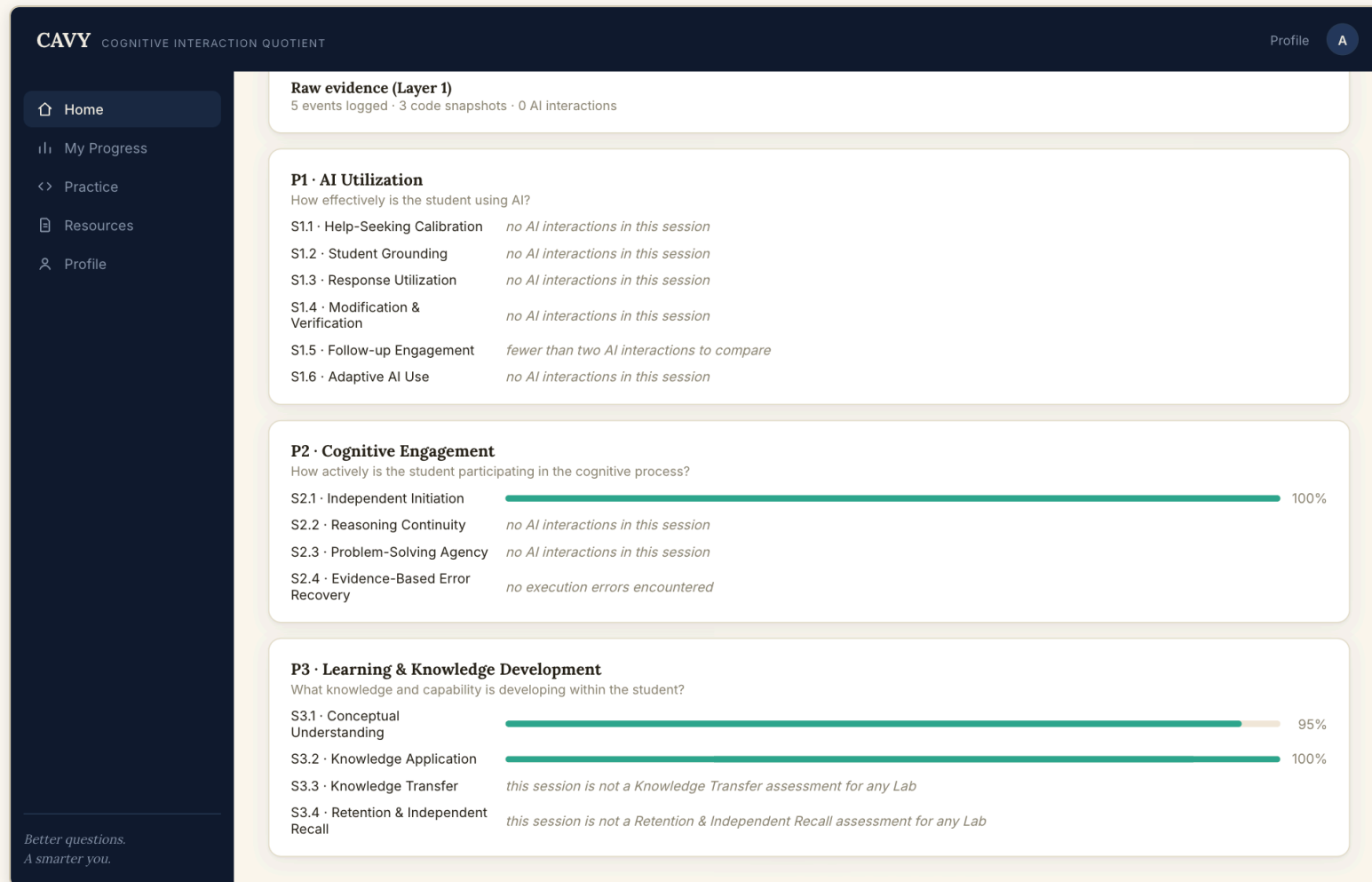
After the assessment the student explains their solution in their own words. That explanation feeds the Conceptual Understanding signal.

# Submission confirmation (Assessment)



Submitted. The teacher now sees this student's assessment in their report.

# CIQ score (Assessment, with concept check)



The Assessment's score includes what the Learning stage could not: the student's own explanation was assessed (Conceptual Understanding) and the final run's quality.

# 7 • The student's own pages

Resources, profile and progress over time.

# Student Resources

The screenshot displays the CAVY (Cognitive Interaction Quotient) interface for a student. The top header is dark blue with the CAVY logo and the text 'COGNITIVE INTERACTION QUOTIENT' on the left, and a 'Profile' button with a blue circle containing the letter 'A' on the right. A dark blue sidebar on the left contains navigation links: 'Home' (with a house icon), 'My Progress' (with a bar chart icon), 'Practice' (with a code icon), 'Resources' (with a document icon and highlighted in dark blue), and 'Profile' (with a person icon). The main content area has a light beige background. At the top of this area is the title 'Resources' in bold, followed by the text 'Shared with you by Dr. Meera Shah.' and a 'Lab' dropdown menu currently showing 'All resources'. Below this, there are three resource cards. The first card, titled 'Before you start' with an 'Instructions' tag, says 'Read this first' and 'Loops and Sums', and includes a 'Read' button. The second card, titled 'Python tutorial: for loops' with a 'Link' tag, shows the URL 'https://docs.python.org/3/tutorial/controlflow.html#for-statements' and includes an 'Open link' button. The third card, titled 'Loops-cheat-sheet' with a 'File' tag, describes it as 'A one-page summary of loop syntax', shows 'Loops-cheat-sheet.pdf · 17 KB', and includes 'Open' and 'Save as...' buttons. All cards indicate they are 'From Dr. Meera Shah · 03/10/2026'. At the bottom of the sidebar, the text 'Better questions. A smarter you.' is visible.

**CAVY** COGNITIVE INTERACTION QUOTIENT

Profile **A**

Home

My Progress

Practice

**Resources**

Profile

**Resources**

Shared with you by Dr. Meera Shah.

Lab

All resources

**Instructions** **Before you start**

Read this first

**Loops and Sums**

From Dr. Meera Shah · 03/10/2026

Read

**Link** **Python tutorial: for loops**

<https://docs.python.org/3/tutorial/controlflow.html#for-statements>

From Dr. Meera Shah · 03/10/2026

Open link

**File** **Loops-cheat-sheet**

A one-page summary of loop syntax

Loops-cheat-sheet.pdf · 17 KB

**Loops and Sums**

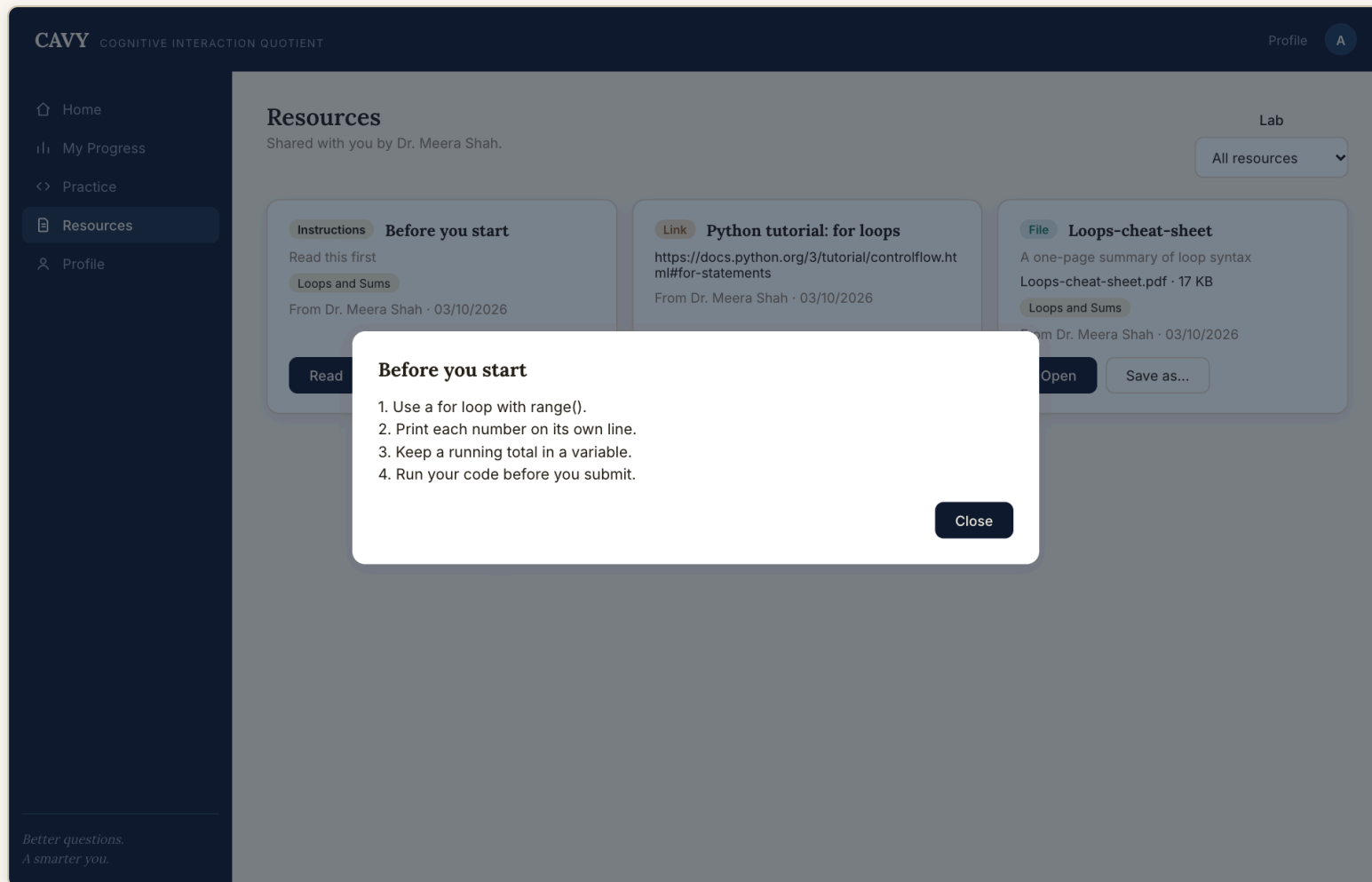
From Dr. Meera Shah · 03/10/2026

Open Save as...

Better questions.  
A smarter you.

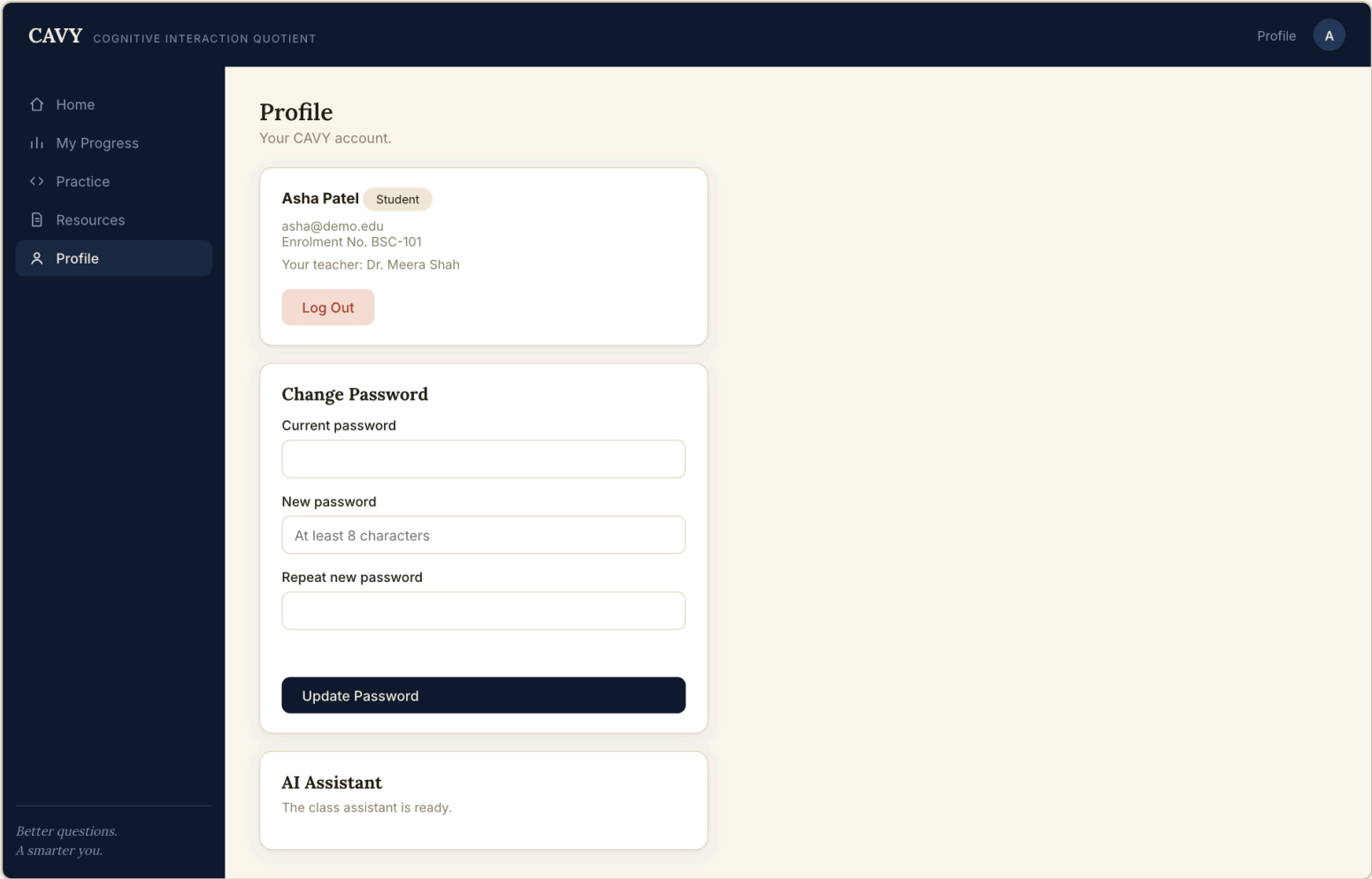
Everything the teacher shared with this student: files (open or save), links, and written instructions. New items appear by themselves.

# Reading written instructions



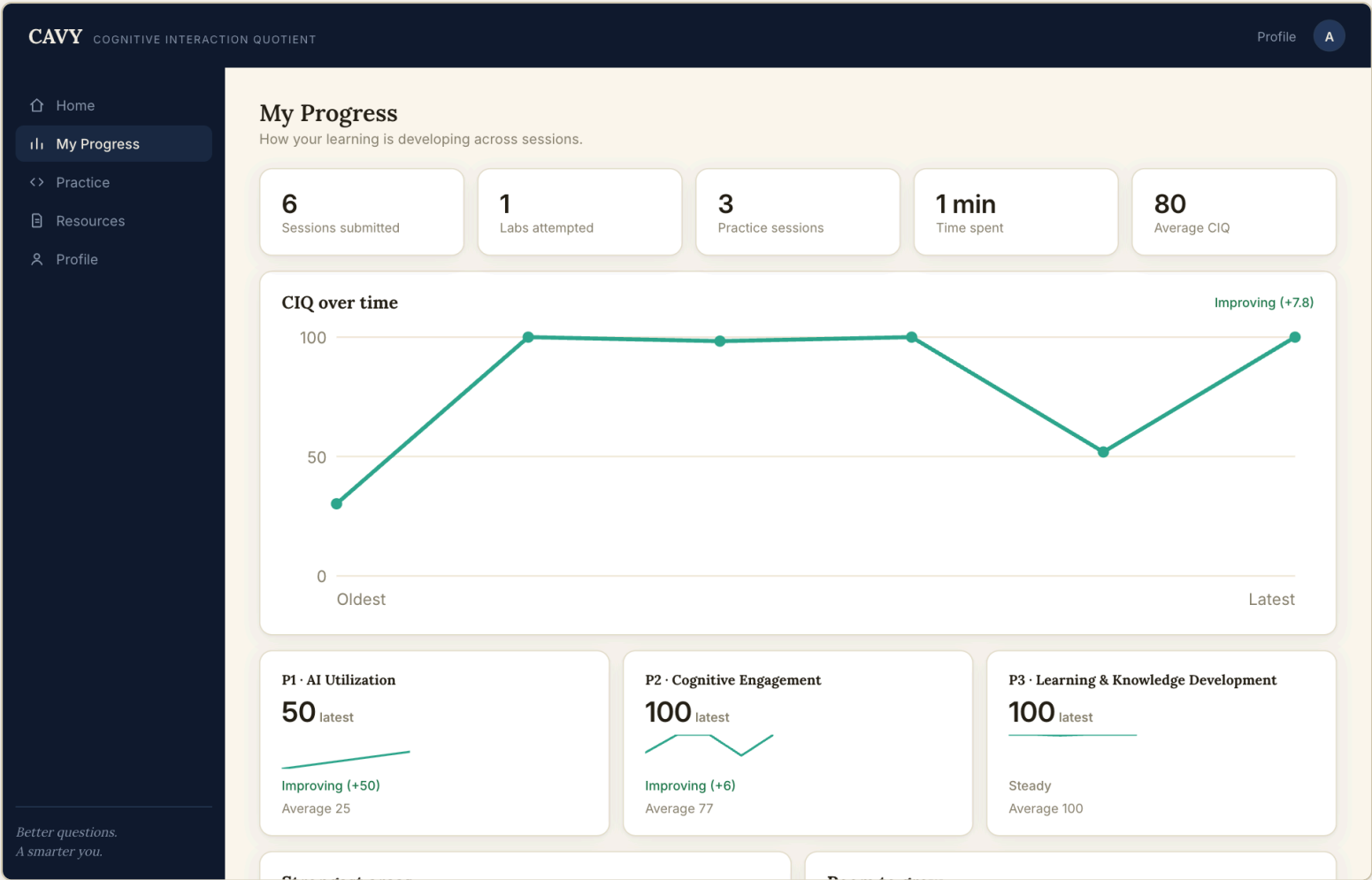
Written instructions open in a reading window.

# Student Profile



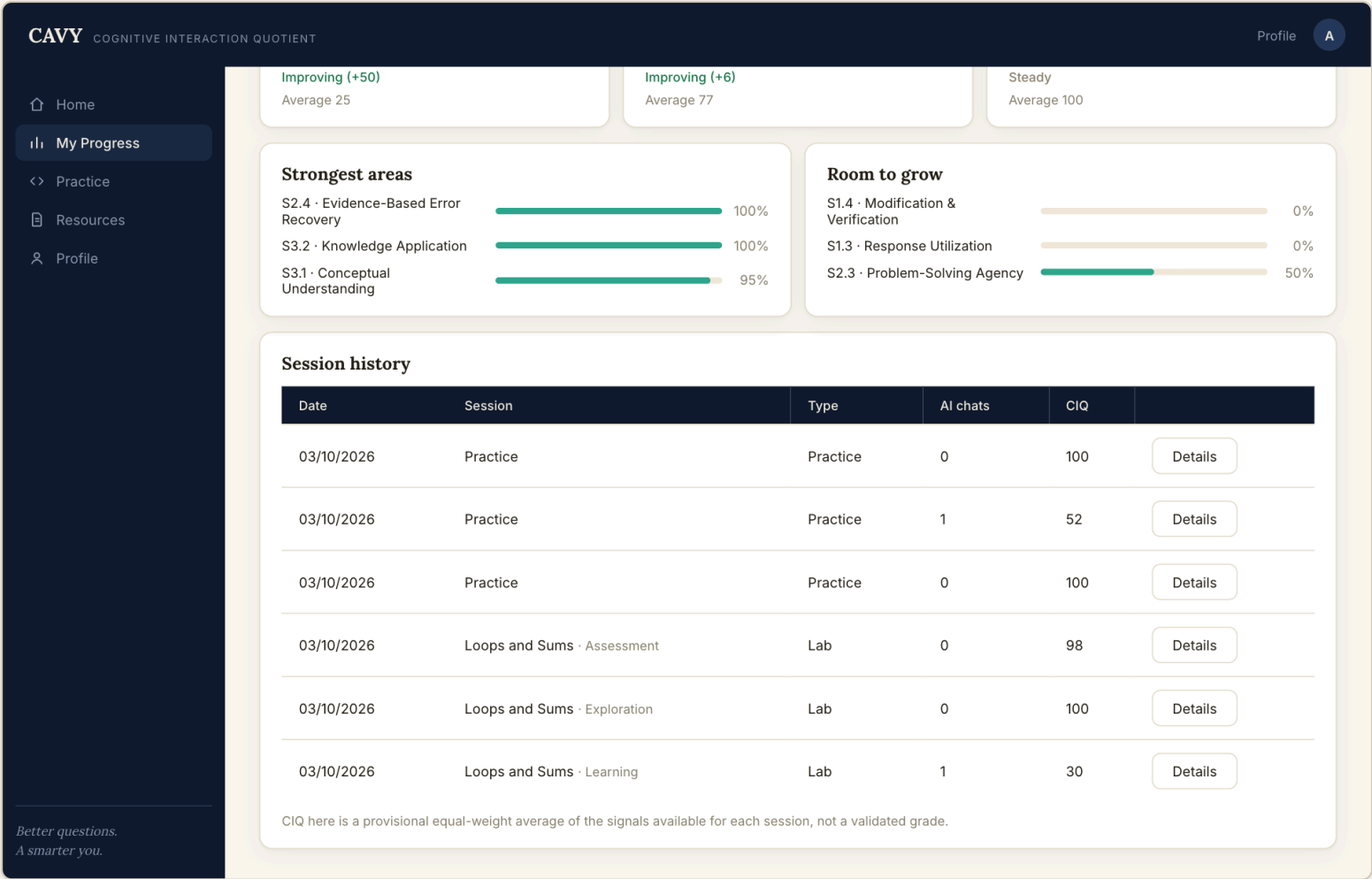
Account details and the student's teacher, a password change form, and the status of the class assistant (students can see it but only teachers change it).

# My Progress (overview)



The student's learning over time: totals, the CIQ trend across sessions, and a card for each of the three pillars.

# My Progress (strengths and history)

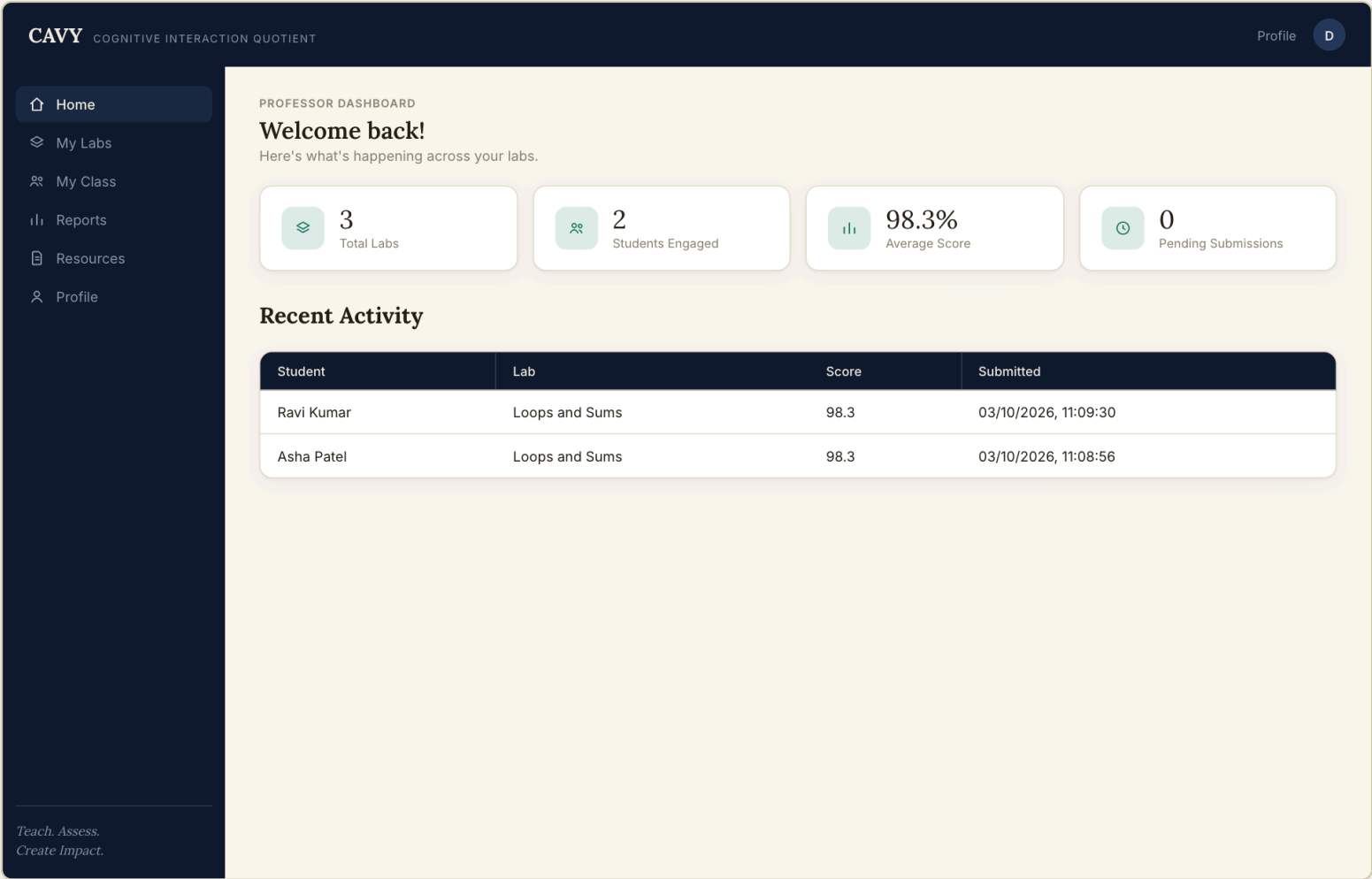


Strongest areas and room to grow, by signal, and a history of every session with a link to its full score.

# 8 · The teacher reviews and manages

Live dashboard, reports, follow-up assessments, editing labs and helping students.

# Teacher Home (dashboard)



The teacher's dashboard (the sidebar's Home): labs, students engaged, average score and pending submissions, with the latest submissions below. It updates by itself the moment a student submits.

# Reports (all labs)

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile

D

Home

My Labs

My Class

Reports

Resources

Profile

REPORTS

Reports

Select a lab to view its report.

Course

Division

Batch

Lab / Assessment

Apply Filter

| Lab                     | Course                 | Division | Batch | Action                 |
|-------------------------|------------------------|----------|-------|------------------------|
| Reversing a Linked List | —                      | —        | —     | <div>View Report</div> |
| Calculating Entropy     | —                      | —        | —     | <div>View Report</div> |
| Loops and Sums          | B.Sc. Computer Science | A        | 2026  | <div>View Report</div> |

Teach. Assess.

Create Impact.

Reports lists every lab; 'View Report' opens one.

# Lab report

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile

D

Home

My Labs

My Class

Reports

Resources

Profile

Back to Dashboard

Loops and Sums

B.Sc. Computer Science | Batch: 2026 | Division: A

Export Report

Loops and Sums

B.Sc. Computer Science

Batch: 2026 · Division: A

2

Total Students

2

Submitted

0

Not Submitted

98.3

Average Score

Course

Division

Batch

Lab / Assessment

B.Sc. Computer Science

A

2026

Loops and Sums

Apply Filter

| Student Name | Enrolment No. | Score | Submitted On         | Status    |
|--------------|---------------|-------|----------------------|-----------|
| Asha Patel   | BSC-101       | 98.3  | 03/10/2026, 11:08:56 | Submitted |
| Ravi Kumar   | BSC-102       | 98.3  | 03/10/2026, 11:09:30 | Submitted |

Follow-Up Assessments

Transfer tasks and retention checks spun off from this Lab.

None yet — add a Transfer Task or Retention Check to collect S3.3/S3.4 evidence.

Add Follow-Up

Teach. Assess.

Create Impact.

A lab's report: each student in the class, whether they submitted, and a provisional overall score. 'Export Report' saves a spreadsheet (CSV) through the computer's Save dialog.

## Add a follow-up assessment (dialog)

The screenshot displays the CAVY Cognitive Interaction Quotient interface. A modal dialog titled "Add Follow-Up Assessment" is open, allowing a teacher to create a new assessment based on an existing one. The dialog includes fields for "Kind" (set to "Transfer Task (a different problem, same concept)"), "Title" (set to "Transfer task: sum the even numbers"), "Problem Description" (set to "Write a program that adds up only the even numbers from 1 to 10."), "AI Assistance" (set to "Restricted"), and "Duration (minutes)" (set to 20). The background interface shows a sidebar with navigation links (Home, My Labs, My Class, Reports, Resources, Profile) and a main content area with a "Loops and Sums" assessment card. The card displays a score of 0 (Not Submitted) and an average score of 98.3. A table below the card lists students and their submission status.

| Student Name | Status    |
|--------------|-----------|
| Asha Patel   | Submitted |
| Ravi Kumar   | Submitted |

A teacher can add a Transfer task (a new problem needing the same idea) or a Retention check (the same idea, days later). They feed the Transfer and Retention signals.

# Edit Session

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile

D

Home

My Labs

My Class

Reports

Resources

Profile

Back to My Labs

Edit Session

Change this lab's details, timing and AI settings.

Session Details

Session Title

\*

Loops and Sums

Course

B.Sc. Computer Science

Topic

Iteration

Division

A

Batch

2026

Difficulty

Easy

▼

Description

Write a program that prints the numbers 1 to 5, one per line, using a for loop. Then extend it to also print their sum.

Learning Workflow

Configure the duration and AI mode for each stage of the session.

Learning

Duration (minutes)

Exploration

Duration (minutes)

Assessment

Duration (minutes)

Teach. Assess.

Create Impact.

Editing a lab. Details, timings and AI levels stay editable; a stage's AI level locks once any student has started it, so earlier evidence keeps its meaning.

## Archive a lab (confirmation)

The screenshot shows the CAVY 'My Labs' interface. A confirmation dialog is displayed over the lab list, asking to archive the lab 'Reversing a Linked List?'. The dialog text states: 'Students will no longer see it and can't start it. Everything already submitted is kept, and you can restore it any time from Show archived labs.' The dialog has 'Cancel' and 'Archive' buttons. The background interface includes a sidebar with navigation links (Home, My Labs, My Class, Reports, Resources, Profile) and a main content area with filters for Course, Division, Batch, and Lab / Assessment. A table of labs is visible, with columns for Lab, Course, Division, Batch, and Action. The lab 'Reversing a Linked List' is highlighted, and its 'Archive' button is visible. Below the table is a '+ Create New Session' button.

CAVY COGNITIVE INTERACTION QUOTIENT

Profile D

MY LABS

**My Labs**  
Manage your labs, view reports, and create new sessions.

Course Division Batch Lab / Assessment

All Courses All Divisions All Batches All Labs Apply Filter

☐ Show archived labs

| Lab                     | Course                 | Division | Batch | Action                                            |
|-------------------------|------------------------|----------|-------|---------------------------------------------------|
| Reversing a Linked List |                        |          |       | <input type="checkbox"/> View Report Edit Archive |
| Calculating Entropy     |                        |          |       | <input type="checkbox"/> View Report Edit Archive |
| Loops and Sums          | B.Sc. Computer Science | A        | 2026  | <input type="checkbox"/> View Report Edit Archive |

+ Create New Session

Teach. Assess.  
Create Impact.

Archiving hides a lab from students and blocks new work, but keeps every submission. It can be restored.

# My Class (three students)

CAVY

COGNITIVE INTERACTION QUOTIENT

Profile

D

Home

My Labs

My Class

Reports

Resources

Profile

My class

The students in your class. Only you (and an administrator) can manage them or share resources with them.

| Name       | Email         | Enrolment No. | Action                                      |
|------------|---------------|---------------|---------------------------------------------|
| Asha Patel | asha@demo.edu | BSC-101       | <div>Reset password</div> <div>Remove</div> |
| Neha Iyer  | neha@demo.edu | BSC-103       | <div>Reset password</div> <div>Remove</div> |
| Ravi Kumar | ravi@demo.edu | BSC-102       | <div>Reset password</div> <div>Remove</div> |

Add students

Students who have an account but aren't in anyone's class yet. A student can be in one class at a time.

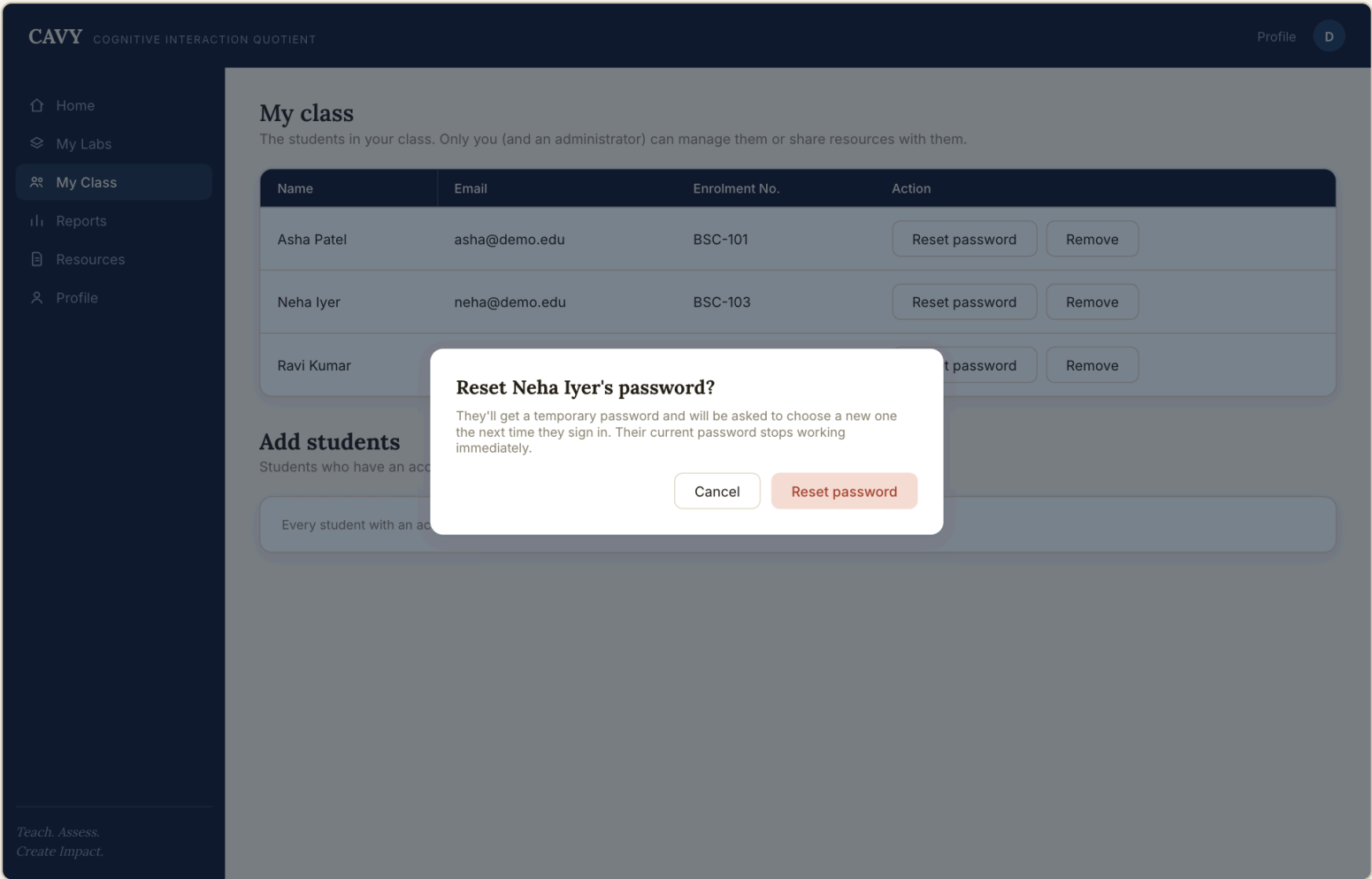
Every student with an account is already in a class. New students appear here after they sign up.

Teach. Assess.

Create Impact.

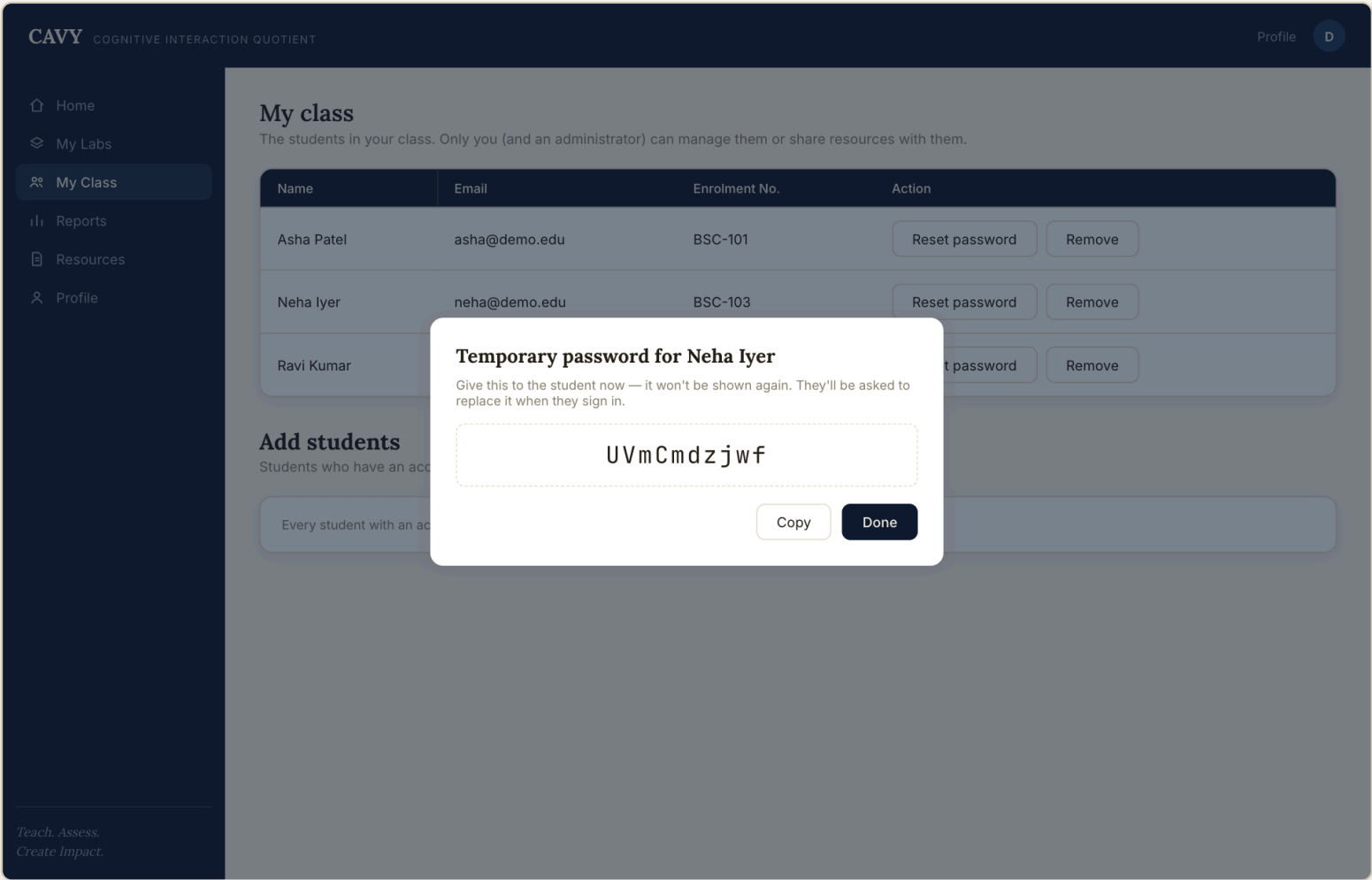
The class table. Only this teacher (and an administrator) can manage these students or share resources with them.

# Reset a student's password (confirmation)



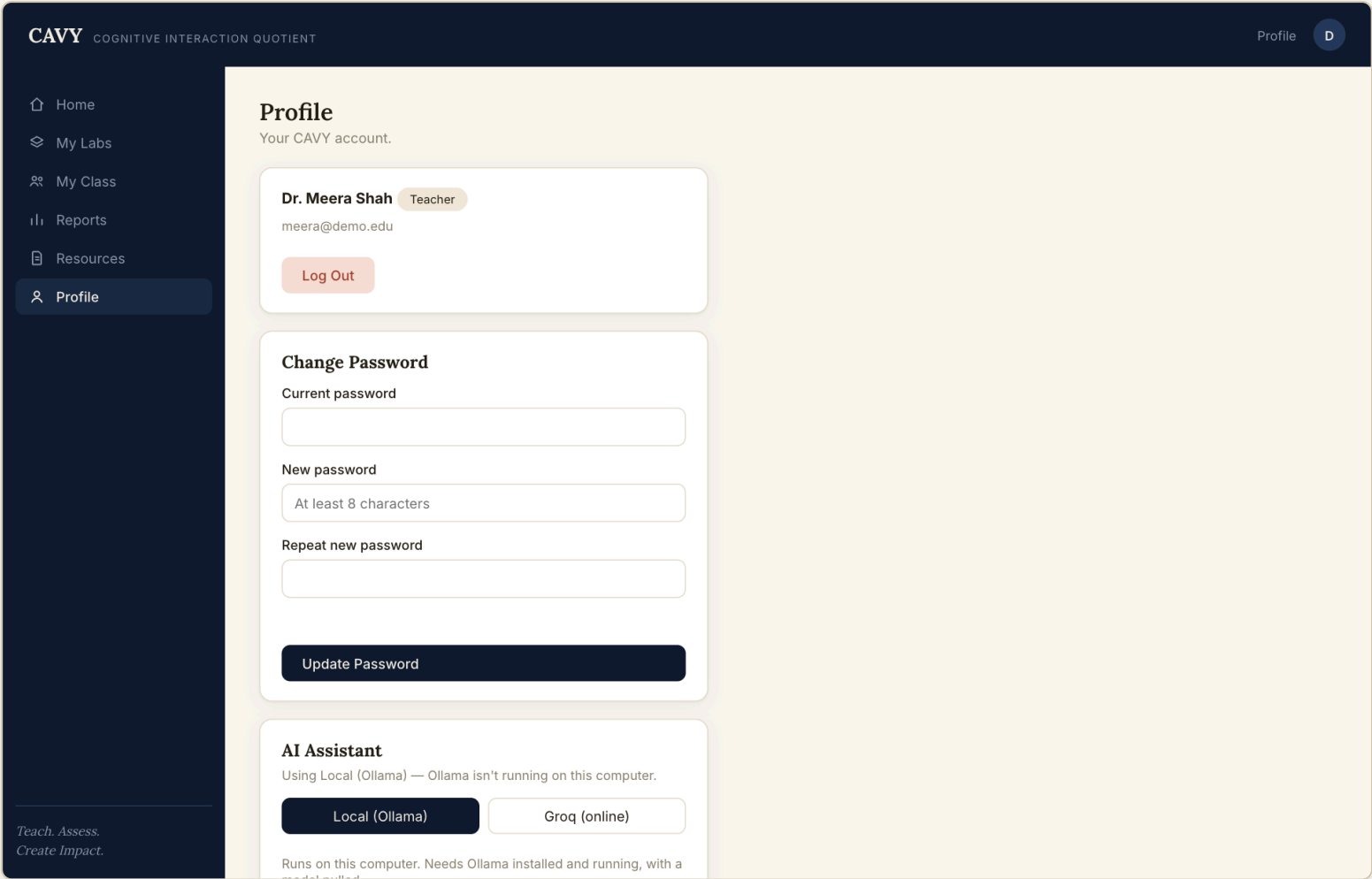
A teacher can help a locked-out student. The old password stops working immediately.

# Temporary password (shown once)



The student gets a temporary password, shown only now. They must choose their own at next sign-in.

# Teacher Profile

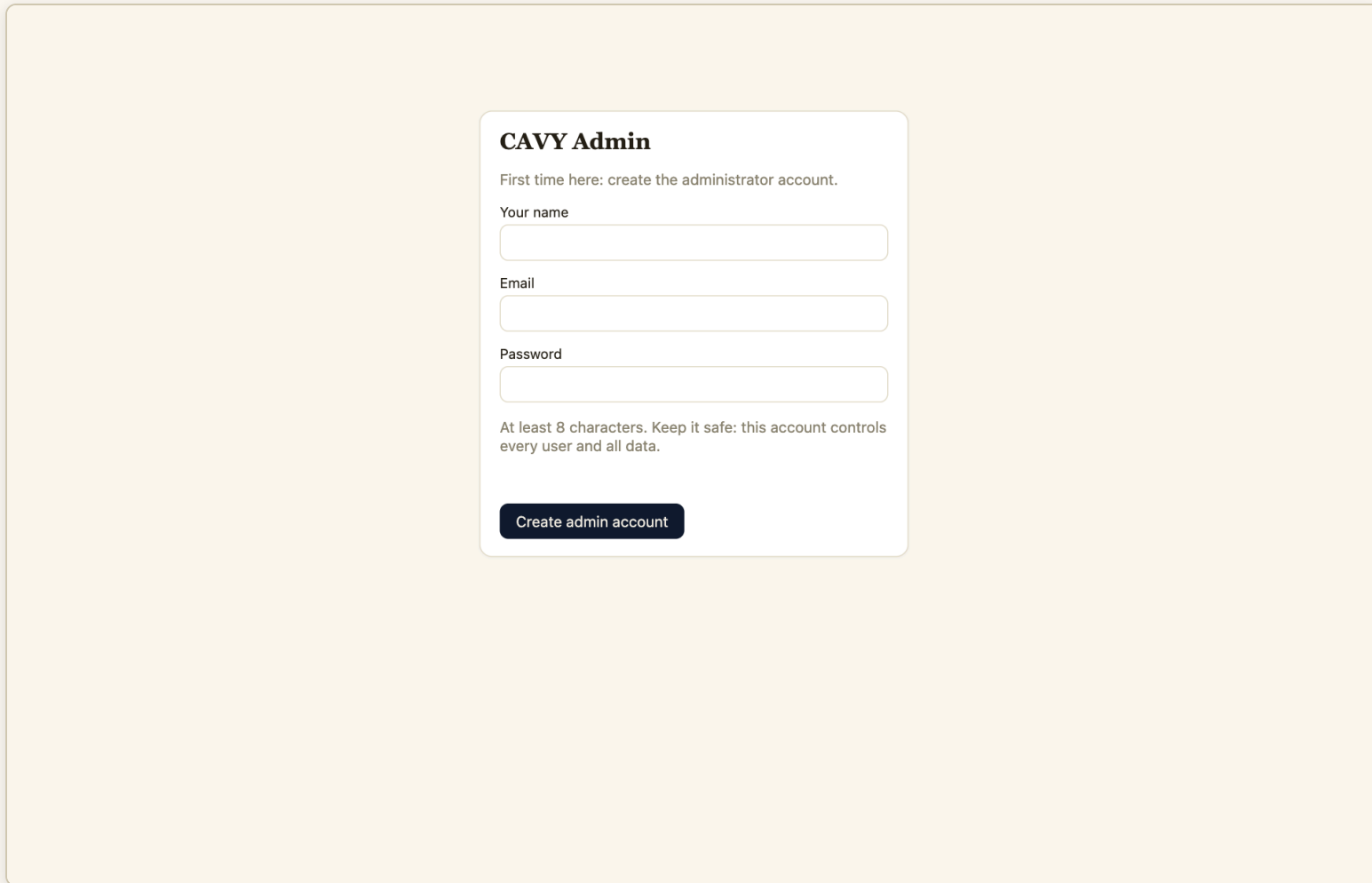


Account details, password change, and the class-wide AI assistant settings (only teachers and administrators can change them).

# 9 · The administrator panel

Everything the person running the server can see and control.

## Admin: first-time setup

A screenshot of a web form titled "CAVY Admin" for creating an administrator account. The form is centered on a light beige background. It includes input fields for "Your name", "Email", and "Password". Below the password field is a warning message: "At least 8 characters. Keep it safe: this account controls every user and all data." At the bottom is a dark blue button labeled "Create admin account".

**CAVY Admin**

First time here: create the administrator account.

Your name

Email

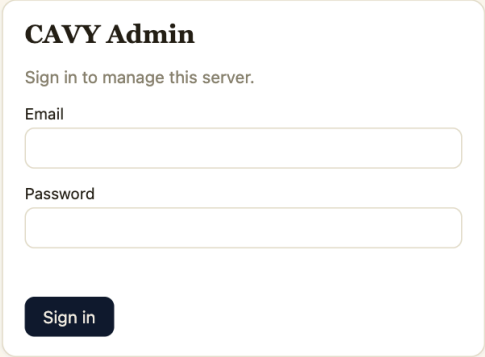
Password

At least 8 characters. Keep it safe: this account controls every user and all data.

Create admin account

The first time the server is opened in a browser, it asks for an administrator account. That account controls every user and all data.

## Admin: sign in



The image shows a sign-in form for the CAVY Admin interface. The form is centered on a light beige background. It has a white title 'CAVY Admin' and a subtitle 'Sign in to manage this server.' Below these are two input fields: 'Email' and 'Password'. At the bottom is a dark blue 'Sign in' button.

**CAVY Admin**

Sign in to manage this server.

Email

Password

Sign in

After setup, the administrator signs in with email and password. Repeated wrong passwords lock the account for a few minutes.

# Admin: Overview (live)

CAVY

ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Overview

Live: updates itself every few seconds.

5

Accounts

3 students · 2 professors

3

Signed in now

1

Working on a task now

0

Disabled accounts

4

Labs

10

Sessions

9 submitted

52

Events logged

2

AI chats

Signed in now

Sign everyone out

| Name           | Role      | From      | Signed in            | Last active |          |
|----------------|-----------|-----------|----------------------|-------------|----------|
| Dr. Meera Shah | professor | 127.0.0.1 | 03/10/2026, 16:40:05 | 3s ago      | Sign out |
| Asha Patel     | student   | 127.0.0.1 | 03/10/2026, 16:40:06 | 2s ago      | Sign out |
| Ravi Kumar     | student   | 127.0.0.1 | 03/10/2026, 16:40:07 | 0s ago      | Sign out |

Working on a task right now

| Student    | Task     | Stage    | Started              | Last activity | Events | AI chats |                      |
|------------|----------|----------|----------------------|---------------|--------|----------|----------------------|
| Asha Patel | Practice | Practice | 03/10/2026, 16:39:32 | 36s ago       | 1      | 0        | <a href="#">Open</a> |

Latest activity

AI assistant

Live overview that refreshes itself: accounts, who is signed in right now (and from which computer), who is mid-task, and the latest activity. Each person can be signed out with one click.

# Admin: Overview (activity)

CAVY

ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Overview

Live: updates itself every few seconds.

5

Accounts

3 students · 2 professors

3

Signed in now

1

Working on a task now

0

Disabled accounts

4

Labs

10

Sessions

9 submitted

52

Events logged

2

AI chats

Signed in now

Sign everyone out

| Name           | Role      | From      | Signed in            | Last active |          |
|----------------|-----------|-----------|----------------------|-------------|----------|
| Dr. Meera Shah | professor | 127.0.0.1 | 03/10/2026, 16:40:05 | 3s ago      | Sign out |
| Asha Patel     | student   | 127.0.0.1 | 03/10/2026, 16:40:06 | 2s ago      | Sign out |
| Ravi Kumar     | student   | 127.0.0.1 | 03/10/2026, 16:40:07 | 0s ago      | Sign out |

Working on a task right now

| Student    | Task     | Stage    | Started              | Last activity | Events | AI chats |                      |
|------------|----------|----------|----------------------|---------------|--------|----------|----------------------|
| Asha Patel | Practice | Practice | 03/10/2026, 16:39:32 | 36s ago       | 1      | 0        | <a href="#">Open</a> |

Latest activity

AI assistant

Further down: who is working on which task right now, the latest events, the AI assistant's status and the latest sessions.

# Admin: Users

CAVY

ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Priya Nair (Administrator)

Sign out

Users

3 students · 2 professors. Changing an email or password signs that person out. Disabling keeps their history but blocks sign-in. A student is in one teacher's class; only that teacher (or you) can share resources with them or manage them.

Add account

Professors (2)

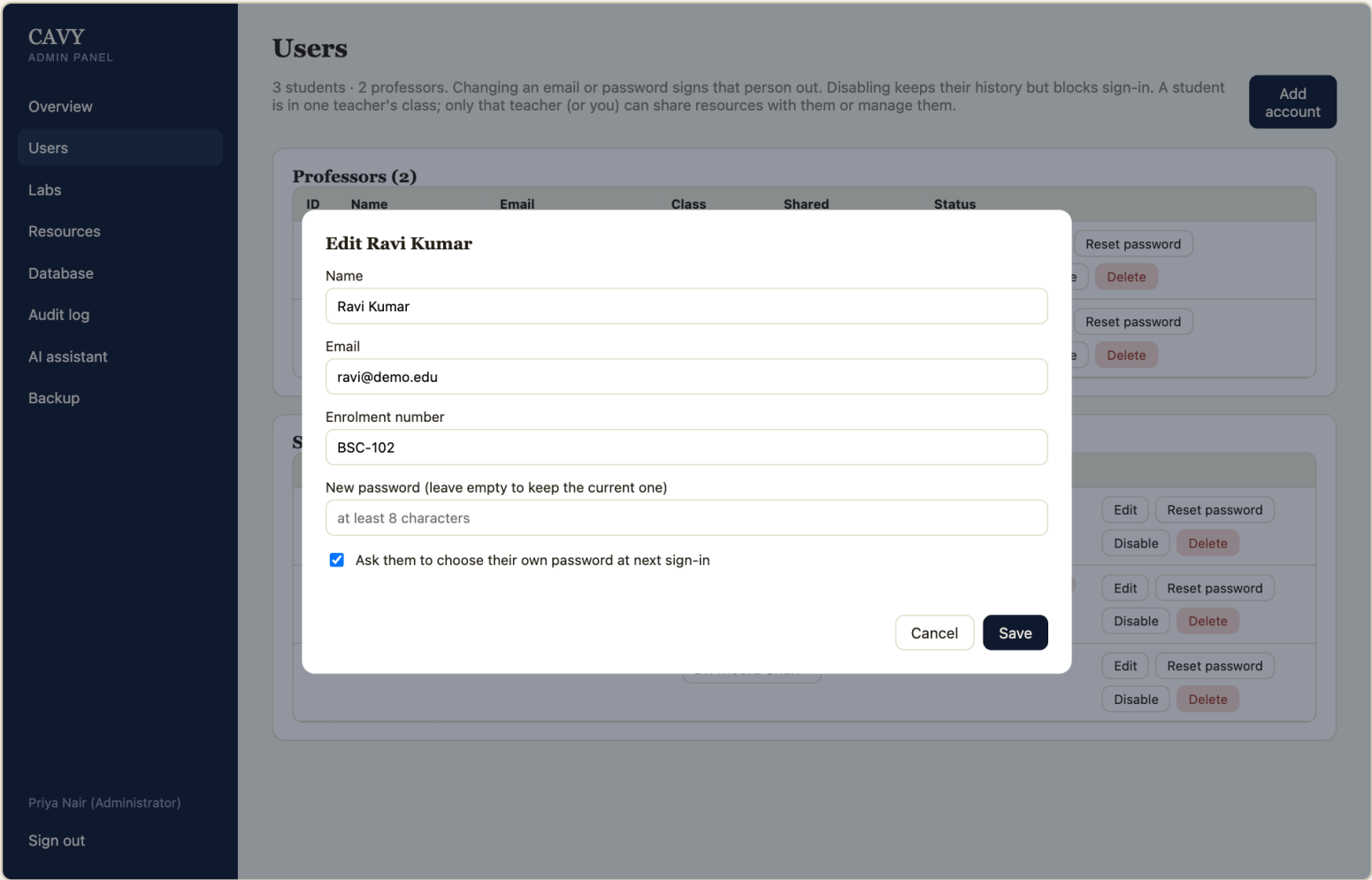
| ID | Name           | Email          | Class      | Shared      | Status |                                            |
|----|----------------|----------------|------------|-------------|--------|--------------------------------------------|
| 1  | Dr. Meera Shah | meera@demo.edu | 3 students | 3 resources | online | <div>EditReset passwordDisableDelete</div> |
| 2  | Dr. Arjun Rao  | arjun@demo.edu | 0 students | 0 resources |        | <div>EditReset passwordDisableDelete</div> |

Students (3)

| ID | Name       | Email         | Enrolment | Teacher        | Sessions | Status               |                                            |
|----|------------|---------------|-----------|----------------|----------|----------------------|--------------------------------------------|
| 1  | Ravi Kumar | ravi@demo.edu | BSC-102   | Dr. Meera Shah | 3        | online               | <div>EditReset passwordDisableDelete</div> |
| 2  | Neha Iyer  | neha@demo.edu | BSC-103   | Dr. Meera Shah | 0        | must change password | <div>EditReset passwordDisableDelete</div> |
| 3  | Asha Patel | asha@demo.edu | BSC-101   | Dr. Meera Shah | 7        | online               | <div>EditReset passwordDisableDelete</div> |

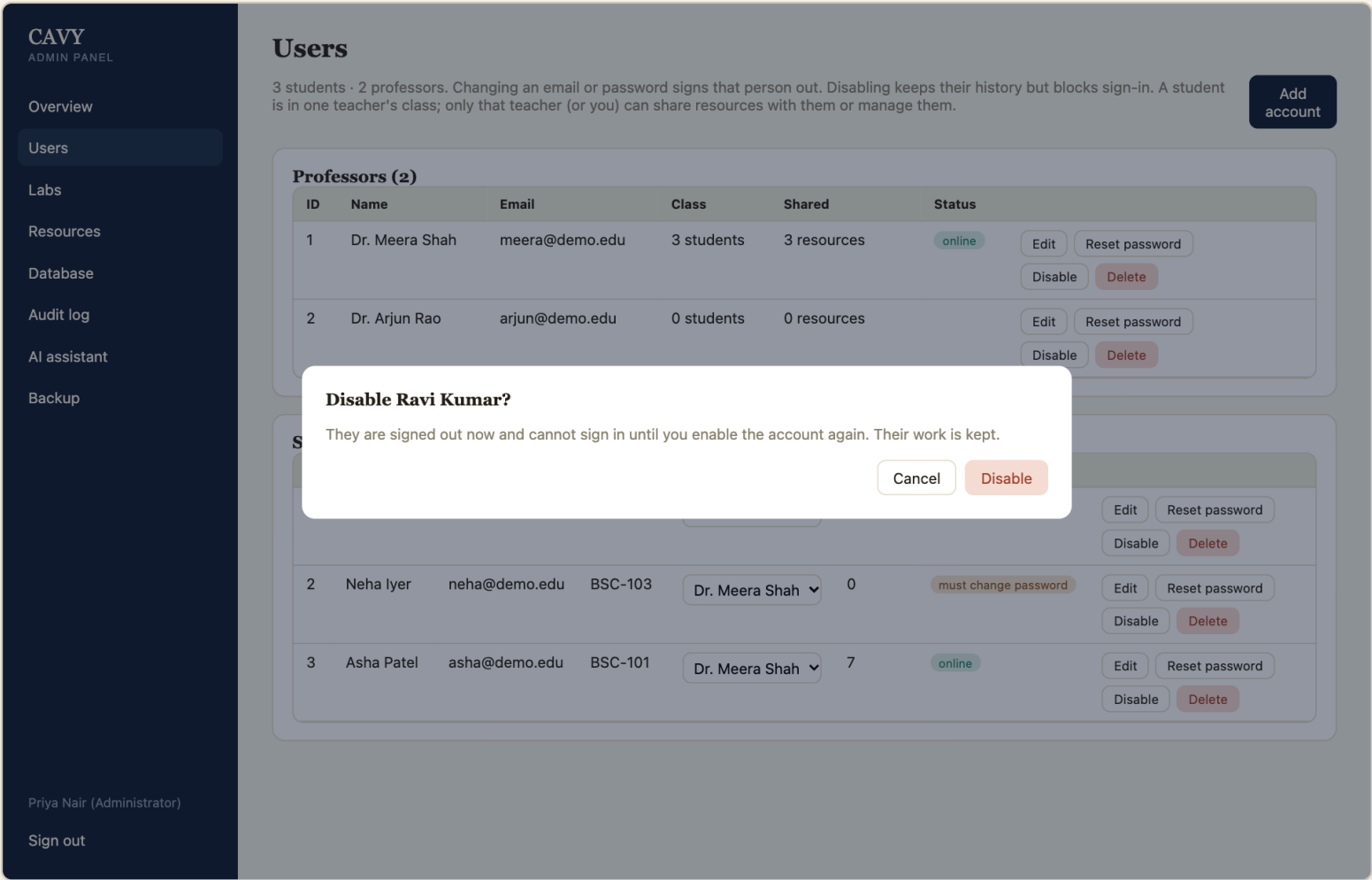
Every account, with online status and, for students, which teacher's class they are in (changeable from the dropdown). Accounts can be edited, reset, disabled or deleted.

# Admin: Edit an account



Change a name, email, enrolment number or password. Changing sign-in details signs the person out at once.

# Admin: Disable an account (confirmation)



Disabling keeps all history but blocks sign-in and signs the person out immediately.

# Admin: Labs

CAVY  
ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Priya Nair (Administrator)

Sign out

Labs

The tests being run, and how many students are working on or have finished each. Archiving hides a lab from students; nothing is deleted.

| ID | Title                   | Professor      | Course                 | Difficulty | Students | In progress | Submitted | Status |         |
|----|-------------------------|----------------|------------------------|------------|----------|-------------|-----------|--------|---------|
| 2  | Reversing a Linked List | Dr. Kariya     |                        |            | 0        | 0           | 0         | Active | Archive |
| 3  | Calculating Entropy     | Dr. Kariya     |                        |            | 0        | 0           | 0         | Active | Archive |
| 4  | Loops and Sums          | Dr. Meera Shah | B.Sc. Computer Science | Easy       | 2        | 0           | 6         | Active | Archive |

The tests being run: for each lab, how many students have taken part, how many are working right now, and how many have finished.

# Admin: Resources

CAVY  
ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Priya Nair (Administrator)

Sign out

Resources

Everything professors have shared. Contents stay private to each professor's class; here you can see what exists and remove it.

| ID | Title                      | Type | Detail                                                                                                                | Owner          | Shared with  | Labs | Added      |
|----|----------------------------|------|-----------------------------------------------------------------------------------------------------------------------|----------------|--------------|------|------------|
| 3  | Before you start           | note |                                                                                                                       | Dr. Meera Shah | 1 student(s) | 1    | 03/10/2020 |
| 2  | Python tutorial: for loops | link | <a href="https://docs.python.org/3/tutorial/controlflow.html">https://docs.python.org/3/tutorial/controlflow.h...</a> | Dr. Meera Shah | whole class  | 0    | 03/10/2020 |
| 1  | Loops-cheat-sheet          | file | Loops-cheat-sheet.pdf · 17 KB                                                                                         | Dr. Meera Shah | whole class  | 1    | 03/10/2020 |

Everything teachers have shared, who owns it and who it is shared with. The administrator can remove items.

# Admin: Database browser

CAVY  
ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Priya Nair (Administrator)

Sign out

Database

Read-only. Click a highlighted id to follow it to the related record.

admins1

ai\_interactions2

audit\_log49

code\_snapshots30

concept\_check\_responses2

events52

execution\_results10

professors2

resource\_files1

resource\_labs2

resource\_students1

resources3

sessions10

signal\_scores84

stages9

students3

tasks4

sessions10 rows

Search all columns...

Export

| id ▼ | student_id        | task_id           | stage_id          | started_at          | ended_at | submitted_at               | sync |
|------|-------------------|-------------------|-------------------|---------------------|----------|----------------------------|------|
| 10   | <a href="#">3</a> | <a href="#">1</a> | null              | 2026-10-03T11:09:32 | null     | null                       | null |
| 9    | <a href="#">1</a> | <a href="#">4</a> | <a href="#">9</a> | 2026-10-03T11:09:30 | null     | 2026-10-03T11:09:30.753288 | null |
| 8    | <a href="#">1</a> | <a href="#">4</a> | <a href="#">8</a> | 2026-10-03T11:09:30 | null     | 2026-10-03T11:09:30.745331 | null |
| 7    | <a href="#">1</a> | <a href="#">4</a> | <a href="#">7</a> | 2026-10-03T11:09:30 | null     | 2026-10-03T11:09:30.739712 | null |
| 6    | <a href="#">3</a> | <a href="#">1</a> | null              | 2026-10-03T11:09:30 | null     | 2026-10-03T11:09:30.707535 | null |
| 5    | <a href="#">3</a> | <a href="#">1</a> | null              | 2026-10-03T11:09:05 | null     | 2026-10-03T11:09:30.695305 | null |
| 4    | <a href="#">3</a> | <a href="#">1</a> | null              | 2026-10-03T11:09:05 | null     | 2026-10-03T11:09:05.226657 | null |
| 3    | <a href="#">3</a> | <a href="#">4</a> | <a href="#">9</a> | 2026-10-03T11:08:51 | null     | 2026-10-03T11:08:56.988577 | null |
| 2    | <a href="#">3</a> | <a href="#">4</a> | <a href="#">8</a> | 2026-10-03T11:08:50 | null     | 2026-10-03T11:08:50.951020 | null |
| 1    | <a href="#">3</a> | <a href="#">4</a> | <a href="#">7</a> | 2026-10-03T11:08:12 | null     | 2026-10-03T11:08:45.591419 | null |

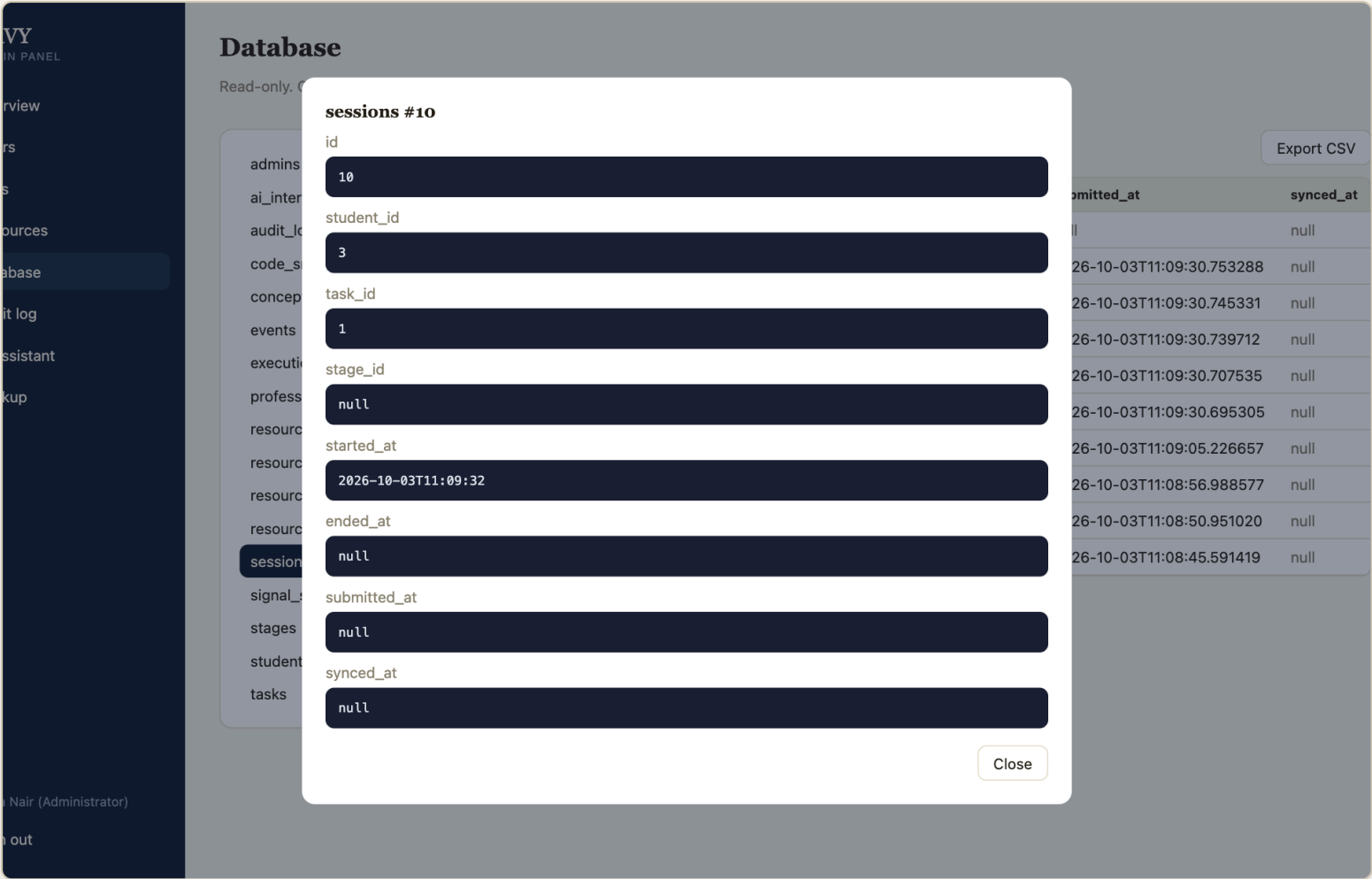
Previous

Page 1 of 1

Next

Read-only access to every table, with search, sorting and CSV export. Password hashes and file contents are never shown. Highlighted ids link to the related record.

# Admin: Database row



Clicking a row shows the whole record, with long text such as code shown in full.

# Admin: One session in detail

CAVY  
ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Priya Nair (Administrator)

Sign out

← Back

Session #10

Asha Patel · Practice · started 03/10/2026, 16:39:32 · not submitted

Signal scores (latest)

Not scored yet.

Event timeline (1)

03/10/2026, 16:39:32 · TASK\_START

AI conversation (0)

The student did not use the AI assistant.

Code runs (0)

No runs.

Latest code

(none saved)

Everything about one student session: signal scores, the event timeline, the AI conversation, every code run and the latest code.

# Admin: Audit log

CAVY  
ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Audit log

Who signed in or out, and what was changed. 49 entries. Passwords and lab content are never recorded here.

Refresh

| When                 | Who                              | Action                 | Detail       | From      |
|----------------------|----------------------------------|------------------------|--------------|-----------|
| 03/10/2026, 16:40:07 | student:Ravi Kumar               | login                  |              | 127.0.0.1 |
| 03/10/2026, 16:40:06 | student:Asha Patel               | login                  |              | 127.0.0.1 |
| 03/10/2026, 16:40:05 | professor:Dr. Meera Shah         | login                  |              | 127.0.0.1 |
| 03/10/2026, 16:40:05 | admin:Priya Nair (Administrator) | force sign out all     | 6 signed out |           |
| 03/10/2026, 16:40:01 | professor:Dr. Meera Shah         | reset student password |              | 127.0.0.1 |
| 03/10/2026, 16:39:59 | professor:Dr. Meera Shah         | add students to class  |              | 127.0.0.1 |
| 03/10/2026, 16:39:59 | professor:Dr. Meera Shah         | login                  |              | 127.0.0.1 |
| 03/10/2026, 16:39:44 | professor:Dr. Meera Shah         | login                  |              | 127.0.0.1 |
| 03/10/2026, 16:39:44 | professor:Dr. Meera Shah         | logout                 |              | 127.0.0.1 |
| 03/10/2026, 16:39:32 | student:Asha Patel               | start practice         |              | 127.0.0.1 |
| 03/10/2026, 16:39:31 | student:Asha Patel               | login                  |              | 127.0.0.1 |
| 03/10/2026, 16:39:31 | student:Asha Patel               | logout                 |              | 127.0.0.1 |
| 03/10/2026, 16:39:30 | student:Ravi Kumar               | submit session         |              | 127.0.0.1 |
| 03/10/2026, 16:39:30 | student:Ravi Kumar               | submit concept check   |              | 127.0.0.1 |
| 03/10/2026, 16:39:30 | student:Ravi Kumar               | start stage            |              | 127.0.0.1 |
| 03/10/2026, 16:39:30 | student:Ravi Kumar               | submit session         |              | 127.0.0.1 |
| 03/10/2026, 16:39:30 | student:Ravi Kumar               | start stage            |              | 127.0.0.1 |
| 03/10/2026, 16:39:30 | student:Ravi Kumar               | submit session         |              | 127.0.0.1 |
| 03/10/2026, 16:39:30 | student:Ravi Kumar               | start stage            |              | 127.0.0.1 |

A record of who signed in or out, and every account, lab and resource change, with the time and computer. Passwords and lab contents are never recorded.

# Admin: AI assistant

CAVY  
ADMIN PANEL

Overview

Users

Labs

Resources

Database

Audit log

AI assistant

Backup

Priya Nair (Administrator)

Sign out

AI assistant

One assistant serves every student and professor. Professors can change it too.

Ready

 Using **ollama** · qwen3:8b

Backend

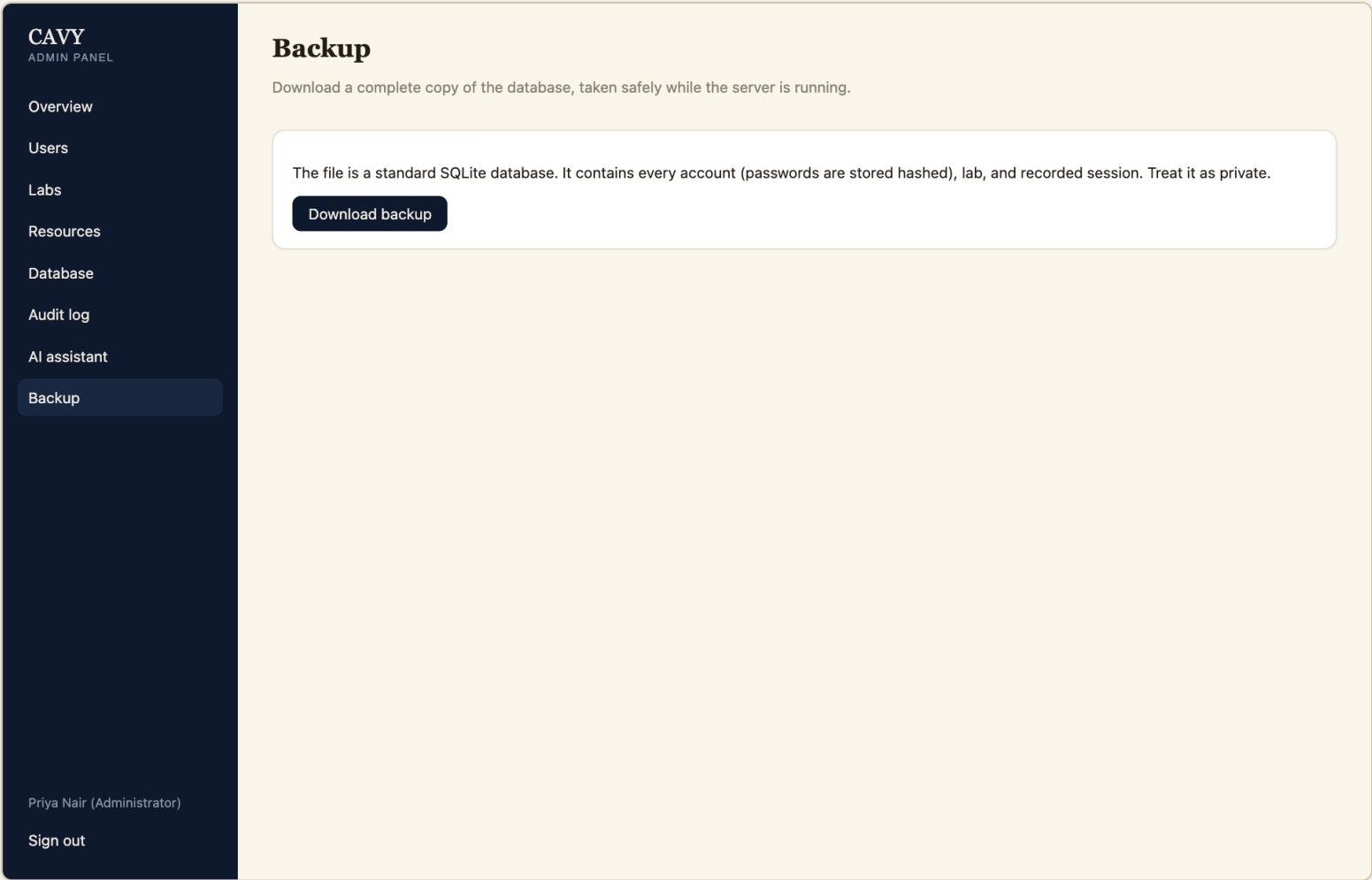
Ollama (runs on this server computer)

The key is kept in the server's memory only and is never written to the database. Restarting the server clears it.

Save and test

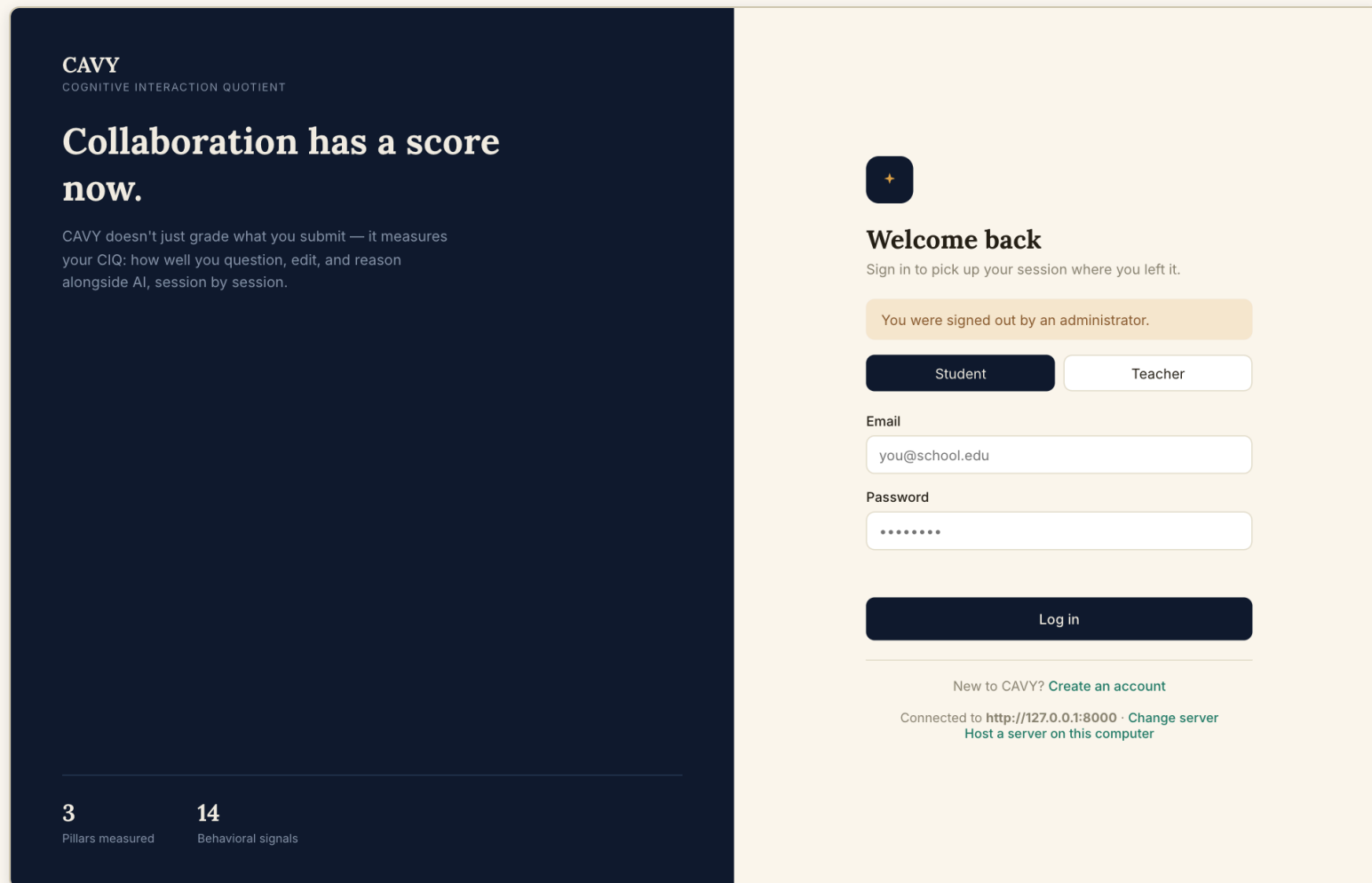
One AI assistant serves the whole class. The administrator (or a teacher) chooses Groq or Ollama here. The key is kept in memory and never written to disk.

# Admin: Backup



One click downloads a complete copy of the database, taken safely while the server keeps running.

## Student signed out by the administrator



When an administrator signs someone out, their app returns to the login screen and says why. No work is lost: everything was already saved on the server.